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Peer Review File

Cross-Scale Observations of Expanding Bifurcated Current Sheets in Fast Magnetic Reconnection

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This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Attachments originally included by the reviewers as part of their assessment can be found at the end of this file.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

Reviewer #2

(Remarks to the Author)
Please see attached

Reviewer #3

(Remarks to the Author)
I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

Reviewer #4

(Remarks to the Author)
The following report has been compiled in accordance with the Nature Communication peer review guidelines.
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[A] Key results
This paper provides a unified interpretation of the bifurcated current sheets observed in laboratory experiments, particle simulations, and satellite observations in the Earth's magnetotail and heliospheric current sheets. In particular, the authors' idea that a narrow layer with large potential gradient due to the two-fluid effect could replace the discontinuous shock surface assumed in the Petschek model can be evaluated as a major step forward in solving the problem of collisionless fast magnetic reconnection. I strongly hope that the authors will improve their manuscript based on the following comments.

[B] Validity
The validity of each data interpretation will be discussed in the following sections, but the results are generally considered to be valid, except for Fig 6. A significant change (sudden bending of the separatrices) is observed in z>95di of Fig. 6a, which is unlikely to be continuous with the 20di<z<80di region. If the distance z~100di has any physical meaning, please explain it. Also, please make sure it is not a problem of the simulation technique (e.g., computational domain).

[C] Significance
The conclusions of this paper bring about a major change in the focus when dealing with magnetic reconnection phenomena close to antiparallel conditions. I believe that their research results will have a significant impact not only on the field of magnetic reconnection research, but also on a broader range of fields including plasma physics, space physics, astrophysics, and nuclear fusion science.

[D] Data and methodology
The manuscript was very easy to understand because the spatial coordinate system was unified for the results of laboratory experiments, simulations, and satellite observations. But please indicate what each variable in the simulation is normalized to. Comments on each physical quantity are provided below.

[D1] Bz in MRX
On line 273, the authors state that there are steep gradients of Bz at x≈±2di in Fig. 3b, but this is not accurate. At least the gradient at x=2di is not steep.

[D2] Jy in MRX
On line 253, it is stated that Jy in the MRX experiment was calculated from the gradient of Bz, but Fig. 3c does not appear to be the x-derivative of Bz in Fig. 3b. Even in flattened regions (x<-2di, -1di<x<0.5di, 2di<x), the gradient of Bz in Fig. 3b is greater than 10% of that in the steep region and is never nearly zero as in Fig. 3c. Furthermore, positive Jy value at x≈-3.5di in Fig. 3c is numerically impossible. Please check the data and processing procedures. Furthermore, MRX results (Fig. 3a) are not very uniform in the z direction at z≈6di. How much does dBx/dz contribute to Jy? Will this result in any significant changes to the interpretation?

[D3] Jy in MMS
On line 394, the authors state that significant Ex and Jy were observed simultaneously, as indicated by the three gray-box regions. The first and last Jy events are thought to be responsible for the step-like changes observed in Bz (lines 351 and 368). However, why does the middle Jy event around 09:49:40 not cause any noticeable change in Bz?

[D4] Potential in MRX
Regarding the potential profile shown in Fig. 3d, the authors state on line 285 that it is "centered at x=0 with sharp drops near x≈±2di", but the plot does not look like that. Also, is there a difference in the potential of both upstream regions? If so, what determines this?

[D5] Ex in MMS
It is stated that the Ex fluctuations observed by MMS shown in Fig. 4d are due to LHDI (line 511). If possible, please provide the results of frequency analysis.

[E] Analytical approach
The data were statistically processed only in the MRX experiment. The results of statistical processing of the data from 250 discharges can be judged to be sufficiently valid.

[F] Suggested improvements
While I do not believe that additional experiments are necessary, it would be very powerful to show data supporting their claim that the formation of bifurcated current sheets strongly depends on plasma β (line 717). References 12 and 17 conduct statistical analyses of the presence or absence of current sheet bifurcations, but how consistent are they with the authors' opinion?

[G] Clarity and context
What does "Expanding" in the title mean? Does it mean constant expansion over time? The data and text of this paper do not show how the structure evolves over time, so it's difficult to imagine what it might look like. Please explain why the authors named it "Expanding."

Regarding "strong in-plane electric field along separatrices" on line 120, I believe the spatial distribution of E is "along" but the direction of E is "across." Please write in a way that avoids misunderstandings.

On line 714, "Simulations in Extended Fig. 3 supports this view" - does this refer to Fig. 8?

[F] References
I have no objections.

[G] Your expertise
N.A.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
The paper has been improved significantly, and most of my concerns have been properly addressed in the revised version. However, several issues still remain and should be further clarified before I can fully support publication.
1. In the MMS event, the authors claim that the current layer Jey is generated by the E×B drift. Since the full 3D electric and magnetic field vectors are available from MMS measurements, it should be possible to directly verify whether Jey can indeed

be explained by Nq(E×B) or not. My concern arises because previous magnetotail observations have shown that the current along the separatrices is often dominated by the parallel component, for example, as shown in Fig. 2e of DOI:10.1002/grl.50537. Moreover, just before the current sheet crossing discussed in this paper, there is another separatrix crossing event in which the current is also dominated by the parallel component (https://doi.org/10.1029/2019GL082538).
2. Lines 290–296: How was the 2D current profile shown in Fig. 3a obtained in the laboratory experiment? From the manuscript, it appears that the current density shown in Fig. 3c was derived from the magnetic field Bz, but the procedure for obtaining the 2D current profile is not sufficiently clear.
3. Lines 328–336: It appears that the density dip is observed on the right-hand side at x=2.0, rather than at x=1.6di. Please check this description carefully.
4. Lines 500–505: The times 08:14:30 and 08:28:15 do not appear to correspond to the sudden increase and sharp decrease described in the text. I could not identify these features at the indicated times.

Reviewer #2

(Remarks to the Author)

Reviewer #3

(Remarks to the Author)
I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

Reviewer #4

(Remarks to the Author)
According to the authors' rebuttal, they responded sincerely to my comments, and their response and revisions to the paper are satisfactory. However, regarding my comment [D3] (comment 4 by reviewer 4 in their rebuttal), although the revised text is provided in the rebuttal, it is not found in the submitted revised manuscript. This appears to be because the authors just forgot to incorporate it into the manuscript, and I would like them to confirm this.

I believe the value of this paper has been further enhanced by the overall revision, particularly the organization of observational results based on beta values.

Version 2:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
All my concerns have been considered. This reviewer has no more comments.

Reviewer #2

(Remarks to the Author)
The authors have answered and explained our questions. We do not have any further questions. The manuscript can be accepted in its present form.

Reviewer #3

(Remarks to the Author)
I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

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Response to Reviewers' Comments

Reviewer 1

We thank the reviewer for valuable comments. The review helped us recognize that several central points of the manuscript, especially the distinction between EBCS, the separatrix region, and the broader exhaust, as well as the intended meaning of "universal" and "quasi-steady", were not presented with sufficient precision. In the revised manuscript, we have therefore clarified the definition of EBCS, distinguished common boundary-layer features from environment-specific kinetic activity, modified several statements, and added a discussion to better explain both the similarities and differences among the laboratory, magnetotail, and heliospheric current-sheet cases.

1. However, the features along separatrix region during reconnection in solar wind and in the magnetosphere are dramatically different. In the magnetosphere, enhanced plasma waves, especially electrostatic waves, strong parallel electric field structure, plasma density peak/cavity, temperature decreases, intense electron beam parallel to the magnetic field, and strong parallel current are generally observed. Of course, the sub-alfvenic ion jets are bounded by the separatrix regions. Some of these features are consistent with EBCS in this paper. On the other hand, in the solar wind, a pair of the slow-mode shocks or discontinuities are detected in the boundary of reconnection-related sub-alfvenic ion jets which has been thought to be quasi-steady reconnection. In this paper, the separatrix region are called EBCS. However, the features along the EBCS in the solar wind are quite different from those in the magnetosphere. For example, in the solar wind, the sharp magnetic field gradient regarded as current layer was detected, without enhanced waves, electron beams etc. So, this reviewer does not think the authors have found many common features along the boundary of the reconnection outflow jets.

   We thank the reviewer for raising this point, which touches upon a central theme of our paper that we now realize requires clearer explanation. We believe that the apparent discrepancy in features between the solar wind (HCS) and the magnetosphere is not due to a fundamental difference in separatrix physics, but is primarily the result of (1) the vast difference in macroscopic system sizes relative to the boundary width, and (2) historical limitations in instrumental temporal resolution.

   First, regarding the 'sharpness' of the boundary and the slow-shock interpretation: In the solar wind (PSP event), the total exhaust width is large (~ $4000d_i$). Therefore, an EBCS boundary that is kinetically wide (~ $20d_i$) appears as an extremely sharp, step-like transition in macroscopic 1D profiles. Because there is such a vast scale separation, this boundary has thus historically been classified purely as an MHD slow-mode shock. In contrast, in the MMS magnetotail event, the entire exhaust width is ~$20d_i$. Consequently, the boundary layers occupy a large fraction of the exhaust, making the gradients appear

less 'sudden' relative to the overall system size. However, our data demonstrate that the underlying two-fluid structure is similar in terms of two-fluid physics.

Second, we agree with the reviewer's insight, supported by Wang et al. (2023), that the central X-line itself can be bursty and turbulent. However, as the outflow propagates, the extended exhaust boundaries (EBCS) maintain a macroscopically coherent structure over large distances. It is precisely this macroscopic stability that has historically led to the 'steady slow-shock' interpretation in the solar wind.

Finally, regarding the perceived absence of enhanced waves, electron beams, and parallel electric fields in the solar wind: This 'absence' is largely an artifact of spacecraft measurement cadences that cannot resolve kinetic-scale fluctuations. When sufficient resolution is available, kinetic separatrix features are indeed observed in the solar wind. For instance, recent Parker Solar Probe observations by Phan et al. (2022) directly identified energetic proton beams leaking along separatrix field lines outside the heliospheric current sheet (HCS). Additionally, high-cadence waveform captures have revealed intense electrostatic waves localized to solar wind current sheets (Malaspina et al., 2013). While the parallel electric field is difficult to resolve directly in the solar wind, its existence is physically required to support the potential well that drives the observed ion acceleration.

A primary conclusion of our paper is that what has historically been interpreted as a macro-scale 'slow-shock' in the solar wind can be unified with magnetospheric and laboratory observations through the two-fluid kinetic framework of EBCS. We have revised the Discussion section to explicitly clarify this scaling effect and include references demonstrating kinetic separatrix features in the solar wind. In particular, we have added the following paragraphs:

"The apparent morphological differences between the boundaries in the solar wind and the magnetosphere can be largely attributed to the vast difference in system sizes. In the heliospheric current sheet, the exhaust width can be extremely large ($\sim 4000~d_i$). Consequently, an EBCS layer of $\sim 20~d_i$ thickness appears as a sharp, step-like discontinuity on macroscopic scales, leading to its historical interpretation as an MHD slow-mode shock. In contrast, in near-Earth environments where the entire exhaust width may only be tens of d_i, the EBCS occupies a much larger relative fraction of the system, making gradients appear less abrupt. Furthermore, a recent observation shows that the localized electron diffusion region in solar wind reconnection can be bursty and turbulent, the resulting macroscopic exhaust boundaries can still propagate as highly coherent, long-lived structures \citep{Wang2023}.

Additionally, while the limited temporal resolution of interplanetary spacecraft has historically obscured micro-physical features like plasma waves and parallel electric fields at solar wind separatrices, high-resolution measurements have confirmed their presence. This includes observations of energetic proton beams on HCS separatrix field lines \citep{Phan2022} and electrostatic waves localized at solar wind current sheets

\citep{Malaspina2013}. Therefore, rather than categorizing solar wind exhausts exclusively as macroscopic fluid shocks, our results suggest that the boundaries of prolonged reconnection across all these environments can be explained by the two-fluid kinetic framework of EBCS."

2. Another main issue is the claim of quasi-steady reconnection. In the magnetosphere, the reconnection both at the magnetopause and in the magnetotail are bursty rather than quasi-steady reconnection, even it can persistent for hours. In PIC simulations, the reconnection rate is generally temporal variation and cannot be steady also. In the solar wind, the ion jets or exhausts are generally called quasi-steady reconnection, since the similar flows were repeatedly observed by different spacecrafts during hours (e.g., Phan et al., nature 2006). However, this explanation is not unique. The steady ion jets do NOT mean the corresponding reconnection is steady also. The reconnection could be still bursty and the resulting outflow jets keep steady during their propagation in the interplanetary space. Recently, based on an MMS observation in the solar wind, it shows that the reconnection responsible for the steady outflows is bursty and turbulent (e.g., Wang et al., Nature Astronomy 2023). So, this reviewer suggests avoid the steady reconnection in the paper, because the reconnection in the magnetosphere and laboratory must be bursty.

   Regarding the excellent recent work by Wang et al. (2023), we agree that the microscopic dynamics at the reconnection X-line (the electron diffusion region) are frequently bursty, temporally varying, and turbulent across the magnetosphere, laboratory plasmas, and the solar wind.

   We agree with the reviewer that our use of the term 'quasi-steady reconnection' required a more precise definition. By 'quasi-steady,' we refer to a system where the macroscopic global reconnection rate (e.g., Comisso and Bhattacharjee 2016) remains steady over a prolonged period. While the instantaneous electric field at the local X-line can be highly dynamic or bursty (even in our controlled MRX experiments) the time-averaged global reconnection rate is continuous. It is this global steadiness that allows macroscopic boundary structures like the EBCS to persist. We have clarified this critical distinction between local X-line burstiness and global quasi-steadiness in the revised text.

   Crucially, the Wang et al. (2023) paper perfectly encapsulates the very physical picture we are conveying. While they observe a highly turbulent diffusion region, they explicitly note that "Far away from the turbulent reconnection site, a Petschek-type exhaust bounded by the slow-mode shock was detected." This observation perfectly aligns with and supports our conclusion: even if the localized X-line is bursty, the boundaries organizing the extended exhaust (which fluid models interpret as slow-mode shocks and our kinetic framework identifies as EBCS) act as robust, macroscopically stable structures.

   Regarding the rarity of EBCS in the magnetosphere compared to the solar wind: As noted in our original manuscript, we attribute this to the stability of global boundary

conditions and the plasma β. The global conditions of the magnetosphere change much more rapidly than those of the heliospheric current sheet (HCS), making it much rarer to achieve and maintain the prolonged quasi-steady conditions required for EBCS. Furthermore, in low-β regimes typical of many magnetotail events, the current sheet is highly susceptible to fragmentation via plasmoid formation or the lower hybrid drift instability (LHDI), which destroys the EBCS structure. We have revised the Introduction and Discussion sections to make these definitions and environmental distinctions much clearer.

In Introduction, we added:
“Here, we use the term `quasi-steady' to describe reconnection where the global reconnection rate \citep{Comisso2016}, when averaged over several Alfv\'en times, remains relatively constant over a prolonged period. While high-resolution observations and simulations show that the instantaneous, electron-scale dynamics at the central X-line are frequently bursty and turbulent, the resulting macroscopic exhaust structures can persist over large temporal and spatial scales under the right global conditions. Indeed, a recent space observation shows that even when the central diffusion region is highly turbulent, a stable Petschek-type exhaust bounded by sharp discontinuities can persist far downstream \citep{Wang2023}. This study focuses on these long-lived, macroscopically stable boundary structures like EBCS that organize these extended exhaust regions.”

We have revised the second to the last paragraph of Discussion:
“The infrequent observation of EBCS in the near-Earth magnetotail can be attributed to the rapidly changing global boundary conditions of the magnetosphere, which rarely allow for prolonged quasi-steady flux annihilation. This scarcity aligns precisely with statistical surveys characterizing bifurcated current sheets as relatively rare or ``atypical” in the magnetotail \citep{Asano2005}. In contrast, the global conditions of the heliospheric current sheet evolve much more slowly, allowing macroscopic boundary structures like those observed by PSP to persist, a physical reality supported by the statistically frequent observation of bifurcated sheets in the solar wind \citep{Gosling2008}. Furthermore, this steadiness is strongly mediated by the plasma β. In moderate β environments (such as MRX and the PSP event), global quasi-steadiness is maintained. While exploring a comprehensive range of β in MRX is challenging due to the limited accessible parameter space, making this an objective for next-generation facilities such as FLARE (Facility for Laboratory Reconnection Experiments), we note that MRX discharges lacking EBCS signatures typically has unstable X-lines, reinforcing the strict necessity of quasi-steady dynamics. In low-β environments typical of the magnetotail, the current sheet associated with EBCS is highly susceptible to the lower hybrid drift instability (LHDI) \citep{Le2018}. In addition, large-scale plasmoids are formed, making the reconnection globally bursty (Fig. \ref{fig:lowbeta}), thereby preventing the formation or persistence of coherent EBCS. When global quasi-steadiness is achieved, however, EBCS serve as robust boundary structures regardless of local bursty dynamics at the X-line.”

3. In my view, this paper mainly studied the separatrix region or EBCS. The separatrix region and the exhaust region are frequently confused throughout the paper. The separatrix region should be the region between the separatrices (the first reconnecting magnetic field line) and the reconnected field line. It is a narrow area in the inflow direction and can extend very far along the outflow direction. The potential well and the ion outflow occupy the whole exhaust region not only the separatrix region. In the paper, these two regions should be distinguished evidently. Otherwise, it is different to know what EBCS exactly refers to.

   We thank the reviewer for pointing out this ambiguity. We agree that precise definitions are essential, and we agree with the reviewer's definitions: the 'separatrix region' is the narrow boundary separating reconnecting and reconnected field lines, while the 'exhaust region' is the broader macroscopic area filled with the reconnected plasma outflow.
   The reviewer is correct that the potential well structure and the ion outflow occupy the entire exhaust region. Our intention was to highlight that the sharpest electrostatic potential **drop** (the strong in-plane electric field) occurs strictly localized near the separatrix (typically within a few d_i). It is this localized, strong electric field at the boundary that drives the strong out-of-plane electron ExB drift, thereby forming the ion-scale EBCS specifically near the separatrix region. Consequently, the EBCS acts as the kinetic-scale boundary layer that encloses and feeds the broader exhaust region, where the accelerated ion outflow resides.

   We have systematically reviewed and revised the manuscript to explicitly distinguish between the narrow separatrix region (where the EBCS and potential drops are localized) and the broader exhaust region (which contains the potential well minimum and the bulk ion outflow).

   Current Text: "This potential structure extends along the magnetic field lines, creating significant potential drops along the separatrices that bound the reconnection outflow. Consequently, a strong in-plane electric field, primarily oriented along the inflow direction (E_x), develops near these separatrix regions. Magnetized electrons respond to this electric field by undergoing an $E \times B$ drift…"

   Revised Text: "This potential structure creates a deep potential well that spans the entire macroscopic exhaust region. However, the steep potential drops (and therefore the strongest in-plane electric fields, primarily oriented along the inflow direction, E_x) are highly localized to the narrow separatrix regions bounding the exhaust. Magnetized electrons respond to this localized electric field by undergoing a strong $E \times B$ drift specifically within this boundary layer. This localized drift, coupled with the reconnecting magnetic field component (B_z), is the fundamental mechanism that generates a strong out-of-plane electron current density specifically within the separatrix region, creating the EBCS (orange).

Current Text: "This formation of EBCS and associated electrostatic potential well results in key characteristics of open outflow geometries such as a sudden change in the magnetic field, a sharp increase in ion outflow across the separatrix, and a density enhancement in the exhaust."

Revised Text: "It is important to clearly distinguish the narrow separatrix region from the broader exhaust region it encloses. The formation of the EBCS is localized to the separatrix region, where the sharp potential drop causes a sudden change in the magnetic field and drives a sharp acceleration of ions across the boundary. Once accelerated across this narrow EBCS layer, the resulting bulk ion outflow and density enhancements subsequently fill the entire macroscopic exhaust region."

4. When comparing the laboratory result, magnetotail observation, and the solar wind observation, in my view, the magnetic field Bz among all three events are quite different. In the solar wind, the Bz gradient at the two vertical dashed lines are very steep. But, the claimed steed Bz gradient in Figs 3 and 4 is much weaker. In the magnetotail (Fig. 4), the current density is not two peaks but at least four peaks, thus the bifurcated current sheet should be inaccurate. The normal electric field Ex in the Fig. 4 was negative in the left half of the current sheet and mainly located in the separatrix region. It seems Ex in the following half was not always positive as claimed in the paper. The electron flow was not shown here. Is it consistent with E_cross_B here? How did the authors get the coordinates in Fig. 4? It is largely different from GSE/GSM coordinates. About the solar wind event, did the authors check the Rankine–Hugoniot relations at the sharp boundary ?

We thank the reviewer for their careful examination of the data in Figures 3, 4, and 5. The reviewer raises several distinct and important points regarding the local measurements, which we address individually below:

1) Difference in B_z profiles:
As discussed in our previous response, the apparent difference in the steepness of the B_z profiles between the solar wind (PSP) and the magnetosphere/laboratory events is a scaling effect due to the vast differences in system sizes. In the PSP event, the entire exhaust is ~4000d_i wide, so a boundary layer of ~20d_i appears as a near-vertical step function in macroscopic plots. In the MMS event, the exhaust is only ~20d_i wide total, meaning the boundary layer occupies a much larger relative fraction of the exhaust, making the gradient appear less steep visually. The underlying kinetic thickness of the EBCS, however, is consistent with our two-fluid scaling.

2) Multi-peak current sheet and E_x fluctuations in the MMS event (Fig. 4): The reviewer correctly observes that the MMS event exhibits more complex sub-structures, including minor additional current peaks and fluctuations where E_x is not uniformly positive on the right side. Rather than contradicting the EBCS model, this perfectly illustrates a key point made in our original manuscript: the EBCS is highly susceptible to further bifurcation and fragmentation into multi-layer structures when the plasma beta is

low. Among our three examples, the MMS event has the lowest electron beta (~0.17). In such low-beta regimes, secondary instabilities such as the lower hybrid drift instability (LHDI) are heavily favored, which causes the potential structure (E_x) to fluctuate and the current sheet to fragment. We have added text to Section 'Spacecraft observation of EBCS in Earth's Magnetotail' to explicitly clarify that these minor peaks and E_x fluctuations are expected signatures of secondary instabilities operating on the primary EBCS structure.

We have made the following change:
Current Text: "Together, these observations indicate that complex, multi-layered current structures can develop near the separatrix during collisionless reconnection."
Revised Text: "Together, these observations indicate that complex, multi-layered current structures can develop near the separatrix. As the MMS event features the lowest electron beta ($\beta_e \approx 0.17$) of our three examples, the current sheet is highly susceptible to secondary instabilities, such as the lower hybrid drift instability (LHDI). This results in the observed fluctuations in E_x and the further fragmentation of the EBCS into the minor secondary peaks visible in Fig. 4e. This demonstrates that while the EBCS is the primary organizing boundary, it can exhibit multi-layer sub-structures in low-β environments."

3) Electron flow and ExB consistency: Regarding the electron flow, we show the out-of-plane electron current density J_{ey} (which is essentially -e n_e V_{ey}) in Fig. 4e. As noted in the text, the enhancements in J_{ey} tightly coincide with the localized intervals of strong E_x, which confirms that the out-of-plane electron flow is indeed primarily driven by the ExB drift, consistent with our theoretical framework.

4) Coordinate system for Fig. 4:
The coordinates used in Figure 4 are a local boundary-normal (xyz / LMN) coordinate system, not standard GSE/GSM. As detailed in the 'Methods' section of the original manuscript (under 'Coordinate transformation for PSP and MMS events'), these coordinates were obtained via Minimum Variance Analysis (MVA) of the magnetic field over the interval 09:46:30–09:52:00. We have updated the Figure 4 caption to direct readers to the Methods section for the exact transformation matrix.

We have made the following change:
Current Text: "a, Magnetic field (($\mathbf{B}$) in the local magnetic coordinate system."
Revised Text: "a, Magnetic field (($\mathbf{B}$) in the local boundary-normal coordinate system (determined via Minimum Variance Analysis; see Methods for the precise transformation from GSE coordinates)."

5) Rankine-Hugoniot (RH) relations for the solar wind event:
We sincerely thank the reviewer for this suggestion. Prompted by your comment, we evaluated the macroscopic 1D MHD jump conditions across the sharp boundaries in the

PSP event. We found that strict 1D Rankine-Hugoniot (RH) relations (such as 1D mass flux conservation and static pressure balance) are not perfectly satisfied across these specific boundaries.

However, this discrepancy is not a violation of fundamental conservation laws, but rather a reflection of the inherently two-dimensional, dynamic nature of the reconnection exhaust. The apparent failure of 1D mass conservation (where density increases without a proportional drop in normal velocity) is naturally explained by 2D mass injection from the X-line flowing along the separatrix. Similarly, static macroscopic force imbalances are accounted for by active plasma acceleration and deceleration within the exhaust.

Because these 2D dynamic effects make it difficult to accurately treat the boundaries as strictly 1D macroscopic slow-mode shocks, our study focuses specifically on the localized kinetic mechanisms (i.e., EBCS) that dictate the structure of the transition layer itself. We have added text to the solar wind section to explicitly clarify this physical context and why evaluating these boundaries as strictly 1D fluid discontinuities is insufficient.

We have made the following revision:
Current Text: "...sudden density enhancements, and large ion outflows are again evident. Figure 5 shows data from a reconnection event..."
Revised Text: "...sudden density enhancements, and large ion outflows are again evident. Historically, such macroscopic boundaries in the solar wind have often been evaluated using 1D ideal magnetohydrodynamic Rankine Hugoniot conditions to classify them as slow mode shocks. However, because actual reconnection exhausts are inherently two dimensional with expanding separatrices, strict 1D planar conservation laws are rarely perfectly satisfied. In this study, rather than treating these boundaries as slow mode shocks, we focus on resolving the underlying kinetic scale physics: the EBCS structures that fundamentally mediate the transition layer. Figure 5 shows data from a reconnection event..."

5. According to the PIC simulations, in Fig. 2 and Fig. 6, one can find clear evolution of current density and plasma velocity along the z direction. It indicates that the structures in separatrix region should depend on the spatial scale along the outflow direction rather than regardless of the system size claimed in the paper.

   We thank the reviewer for their careful attention to the simulation results in Figures 2 and 6. The reviewer correctly notes that the current density and plasma velocity profiles look different between these two figures. However, we must clarify that these differences are not due to a breakdown of the EBCS structure at larger spatial scales, but rather because Figure 2 and Figure 6 depict **two distinct simulations run with different physical parameters and initial conditions**.

   As summarized in the table in the 'Additional details on Harris sheet simulations' section

of the Methods , the simulation in Figure 2 uses a physical mass ratio (m_i/m_e = 1836) in a smaller domain (20 x 20 d_i), whereas the simulation in Figure 6 uses a reduced mass ratio (100) in a much larger domain (200 x 40 d_i) to computational feasible track the long-distance persistence of the exhaust. They also feature different initial conditions, background densities, and temperature ratios.

Our claim that the EBCS is a universal structure 'regardless of system size' means that the fundamental cross-sectional boundary physics (the bifurcated electron current, the localized potential drop, and the ExB drift) reliably emerges whether the overarching system is small (like MRX) or massive (like the solar wind). While the macroscopic exhaust cone naturally widens and evolves as plasma flows downstream along the z-direction, the kinetic-scale EBCS boundary structure that bounds this exhaust persists coherently over these extended distances. We have added a brief clarifying sentence to the text when introducing Figure 6 to ensure readers do not confuse the downstream evolution of the exhaust with a scale-dependency of the EBCS formation mechanism.

We have made the following revision:
Current Text: "Figure 6 summarizes the results of this large-domain simulation. The upstream plasma beta is 1.2, representative of PSP observations. Figure 6a show the development of EBCS near the separatrix, extending to the boundary of the simulation domain."

Revised Text: "Figure 6 summarizes the results of this large-domain simulation. The upstream plasma beta is 1.2, representative of PSP observations. Figure 6a shows the development of EBCS near the separatrix. **While the macroscopic exhaust naturally widens and evolves as it propagates downstream along the z-direction, the kinetic EBCS boundary structure persists coherently, extending to the boundary of the simulation domain.**"

6. Lines 77-80: actually, the slow-mode shock has been extensively observed in the solar wind and in the magnetotail also. Its relation to the reconnection is indeed unclear.

We agree with the reviewer that slow-mode shocks are frequently observed in space plasmas. However, in our introduction, we are referring specifically to the classic **Petschek-type switch-off slow shocks** acting as the direct boundaries of collisionless reconnection exhausts. As noted in the text and supported by previous kinetic studies (e.g., Liu et al., 2012), direct evidence for these specific shocks at these exact boundaries remains inconclusive. We have slightly modified the text to clarify this distinction.

Current Text: "While Petschek-like open exhausts have been often inferred from space and laboratory observations under low-resistivity conditions [e.g., 7-9], direct evidence for the predicted switch-off slow shocks has remained inconclusive [10]."

Revised Text: "While Petschek-like open exhausts have been often inferred from space and laboratory observations under low-resistivity conditions [e.g., 7-9], direct evidence for the predicted switch-off slow shocks **directly bounding the collisionless exhaust** has remained inconclusive [10].

7. Line 81: what does the word of “expanding” mean in EBCS? I did not find any discussion on expanding in the paper.

   We thank the reviewer for pointing out this omission. The term 'expanding' refers to the geometric divergence of the two current layers: because they form along the magnetic separatrices, the distance between the bifurcated sheets naturally expands as the separatrices open up further downstream from the X-line. We agree that this should be defined immediately upon first use, and we have updated the Introduction to explicitly clarify this geometric meaning.

   Current Text: "Expanding bifurcated current sheets (EBCS) have been widely reported in heliospheric current sheets [7, 11–15] and Earth’s magnetotail [16, 17]. EBCS consist of double-peaked current layers that extend along magnetic separatrices, accompanied by sharp magnetic field variations, localized electric fields, and enhanced ion outflows."
   Revised Text: "Expanding bifurcated current sheets (EBCS) have been widely reported in heliospheric current sheets [7, 11–15] and Earth’s magnetotail [16, 17]. **These structures are termed 'expanding' because the distance between the double-peaked current layers naturally increases downstream as they follow the diverging separatrices.** They are accompanied by sharp magnetic field variations, localized electric fields, and enhanced ion outflows."

8. Lines 174-181: The potential well centered at x=0 and across the exhaust region has been verified by spacecraft measurement, consistent with the Hall electric field. It is shown in the paper also. But, the significant potential drops along the separatrices haven’t be shown in this paper. In the exhaust region, the electrons should experience E_cross_B drift. In the sepratrix region, the inflowing electron beams along the magnetic field should be there.

   We thank the reviewer for catching this imprecise wording. The reviewer is correct: 'along the separatrices' implies a parallel potential drop driving field-aligned electron beams, which, while present, is not the primary mechanism forming the bifurcated current sheet.

   Our intended meaning was that there is a steep potential drop **across** the separatrix (in the normal/inflow direction), and that this cross-boundary potential structure physically extends downstream along the length of the separatrices. It is this steep normal potential drop that creates the strong in-plane electric field (E_x), which in turn drives the out-of-plane $E \times B$ electron drift to form the EBCS.

We apologize for the confusing phrasing and have corrected the text in the manuscript to explicitly state 'across the separatrices', ensuring the description accurately reflects the perpendicular ExB drift mechanism driving the EBCS.

Current Text: "This potential structure extends along the magnetic field lines, creating significant potential drops along the separatrices that bound the reconnection outflow.
Revised Text: "This potential structure creates a deep potential well that spans the entire macroscopic exhaust region. However, the steep potential drops (and therefore the strongest in-plane electric fields, primarily oriented along the inflow direction, E_x) are highly localized to the narrow separatrix regions bounding the exhaust.

9. Lines 190-206: Where is the sharp increase of the ion outflow Viz ? Fig. 2c shows the ion flow occurs everywhere in the outflow region, and a further enhancement along the separatrix region between z = (2,4). If this enhancement along the separatrix refers to the Petscheck-type exhaust, it is very localized and depended on the z axis, and not a common feature even along the separatrices.

   We thank the reviewer for this observation, which highlights a need for more precise wording in our description of the ion dynamics.

   We agree that the accelerated ion outflow (V_{iz}) physically occupies the entire exhaust region. When we referred to a 'sharp increase', we meant the sharp **spatial gradient** or the **acceleration** of the ions across the boundary, rather than the absolute magnitude of the velocity within the exhaust.

   In a classic Petschek-type configuration, plasma is accelerated as it crosses the localized boundaries (slow shocks in MHD, or EBCS in our two-fluid model). Once the plasma crosses this boundary, it fills the broader exhaust region at this high velocity. Therefore, the sharpest spatial increase in V_{iz} occurs directly across the separatrix layer itself. This localized, cross-boundary acceleration is the defining feature of a Petschek-type exhaust, and it is clearly visible in Figure 2c, where the velocity contours are tightly packed precisely at the separatrix before broadening into the exhaust cavity.

   We have revised the text to explicitly use the terms 'sharp spatial gradient' and 'localized acceleration across the separatrix' to avoid conflating the narrow acceleration region with the broader high-velocity exhaust region.

   Current Text: "Simultaneously, this structure accelerates ions across the separatrix, as shown in Fig. 2c, which plots the ion outflow velocity V_{iz}. A sharp increase in ion outflow is observed just across the separatrix, consistent with the Petschek-type exhaust."

   Revised Text: "Simultaneously, this structure accelerates ions across the separatrix, as shown in Fig. 2c, which plots the ion outflow velocity V_{iz}. **A sharp spatial gradient in the ion outflow velocity, indicating strong, localized acceleration, is observed**

directly across the separatrix layer. Once accelerated across this EBCS boundary, the high-velocity ions fill the broader exhaust region. This localized cross-boundary acceleration is a fundamental kinetic manifestation of a Petschek-type exhaust."

10. Lines 400-405: the temperature data were used to calculate the ion enthalpy flux. However, the temperature is inaccurate in the MMS event, because the thermal ions cannot be covered by the FPI instrument here. This introduces a potentially large uncertainty.

    We thank the reviewer for pointing out the instrumental limitations of the MMS FPI regarding the measurement of thermal ions. We acknowledge that these limitations can introduce uncertainty into the absolute calculation of the ion temperature moments.

    However, we believe that despite this uncertainty, the measured temperature increase and the resulting conclusion that the enthalpy flux dominates the bulk kinetic energy flux makes complete physical sense within the energy budget of this specific event. As shown in our analysis, the potential drop across the separatrix ($e\Phi_0$) is estimated to be greater than 5 kV (see Table 1). Meanwhile, the measured bulk ion outflow is less than 400 km/s, which corresponds to a bulk kinetic energy of less than 1 keV per proton. This massive discrepancy indicates that the majority of the energy gained by the ions as they cross the strong potential drop does not go into bulk acceleration, but instead must be converted into thermal heating and the work required to compress the plasma across the separatrix.

    Therefore, while the exact numerical value of the enthalpy flux may carry some instrumental uncertainty, the physical conclusion that it significantly exceeds the bulk kinetic energy flux is robust and mathematically required by the measured electric fields and bulk flows. We have added a brief discussion of this energy conservation argument to the manuscript to validate the enthalpy flux observation.

    Current Text: "In this event, the enthalpy flux dominates over the bulk kinetic flux, even after subtracting the upstream temperature of 2.8 keV. This indicates that the majority of ion energization takes the form of thermal heating rather than bulk acceleration."

    Revised Text: "In this event, the enthalpy flux dominates over the bulk kinetic flux, even after subtracting the upstream temperature of 2.8 keV. **While instrumental energy-range limitations of the FPI can introduce some uncertainty into the absolute ion temperature moments, this dominance of enthalpy flux is physically required by the overall energy budget. The potential well depth for this event is estimated to exceed 5 kV (Table 1), yet the bulk ion outflow of < 400 km/s accounts for much less than 1 keV of kinetic energy per proton. Consequently, the vast majority of the energy gained by ions traversing the potential drop must be converted into thermal heating and plasma compression.** This indicates that the

majority of ion energization takes the form of thermal heating rather than bulk acceleration."

Reviewer 2 and 3

We would like to sincerely thank Reviewers for their careful reading of our manuscript and for providing such constructive and insightful feedback. Their comments have been highly valuable in helping us refine our physical arguments, clarify our terminology, and strengthen the overall presentation of our cross-scale comparison of Expanding Bifurcated Current Sheets (EBCS).

In the revised manuscript, we have carefully considered and addressed each of the concerns raised. We believe that incorporating the reviewer's suggestions has significantly improved the clarity and impact of our paper. Below, we provide a detailed, point-by-point response to all specific comments, with our replies detailed below and the corresponding additions to the manuscript text clearly indicated.

Major Comments:

1. *The expression for the potential well Φ_0 is given in equation (4) after integrating equation (1) via equations (2) and (3). The integration is performed across the EBCSs. However, the number density, n is considered as average density nav (not described though in the text). The number density changes significantly across the EBCSs as pointed out in most of the figures, in particular Fig. 3e. Why is it feasible to consider an average density across the EBCSs and take it out of the integration?*

   We thank the reviewer for this careful reading of our derivation. The reviewer is correct to point out that the number density changes across the EBCS, and we omit the explicit definition of n_{av} and the justification for this simplification in the original manuscript.

   In this context, n_{av} is defined as the density averaged strictly over the region with a significant current density (i.e., the region where the significant magnetic field change occurs). Treating the density as an average value and pulling it out of the integral is a mathematical simplification made for practical utility. When we calculate the potential drop by integrating the actual, measured density profiles, the results differ from our simple formula by about 10%. The simple formula tends to slightly overestimate the potential well depth because it ignores the diamagnetic current (the electron pressure gradient term) from Equation 1. However, since the primary purpose of Equation 4 is to provide a straightforward observational proxy and to explicitly demonstrate the rough plasma beta dependence (as shown in Equation 5), this 10% difference is acceptable, particularly given the standard uncertainties inherent in both spacecraft and laboratory electric field measurements.

   We have revised the text immediately following Equation 4 to explicitly define n_{av} and to briefly justify this practical simplification.

Current Text: "where Δ represents the change across EBCS. If the field changes sign, magnitudes are added rather than subtracted."

Revised Text: "where Δ represents the change across EBCS. If the field changes sign, magnitudes are added rather than subtracted. **Here n_{av} is the density averaged over the region with a significant current density (i.e., where the significant magnetic field change occurs). Treating the density as an average constant across this specific layer is a practical simplification. Integrating the exact density profile yields values that differ from this simple formula by only $\sim$10\%, as the simplified estimate slightly overestimates the well depth by neglecting the diamagnetic current (electron pressure gradient). Given the typical uncertainties in measuring electric fields and densities, this formula provides a highly reliable, straightforward proxy for the potential drop and clearly highlights the physical scaling.**"

2. *In equation (5), the authors show that the potential is inversely proportional to the upstream β. According to this expression, large β systems have comparatively small potential well and hence the in-plane electric field Ex, which is the main reason for the presence of EBCS due to the E x B drift of electrons. However, they also claim that the EBCSs preferentially occur in higher β plasma systems. This contradiction is crucial to the manuscript's main point and thus should be resolved fully.*

   We are grateful to the reviewer for catching this apparent contradiction. The reviewer's logic regarding Equation 5 is correct: as β becomes very large, the potential well depth becomes increasingly shallow, which suppresses the in-plane electric field and prevents the clear formation of EBCS.

   Our original phrasing of 'moderate to high beta' was imprecise. As the reviewer rightly deduced, EBCS actually require a 'Goldilocks' regime of **moderate β (~ 1)**. If β is too high (e.g., $\beta >> 1$), the potential well is too weak to drive the bifurcated currents. Conversely, if β is too low, the potential well is deep, but the current sheet becomes highly susceptible to fragmentation via plasmoid formation and the lower hybrid drift instability (LHDI), which destroys the coherent EBCS structure (as demonstrated by our low-β PIC simulation).

   The three primary events analyzed in our paper – MRX ($\beta \sim 0.7$), MMS ($\beta \sim 0.6$), and PSP ($\beta \sim 1.2$) – all sit comfortably within this optimal 'moderate β window where the well is strong enough to drive the currents, but the plasma is stable enough to maintain the macroscopic boundaries. We have corrected the terminology throughout the abstract and manuscript from 'moderate to high' to 'moderate β (~ 1)', and we have added a brief explanation of this upper β limit in the discussion of Equation 5 to resolve the contradiction.

   Current Text: "Although EBCS are not present in all reconnection events, they emerge

robustly during quasi-steady antiparallel reconnection, particularly in plasmas with moderate to high beta."

Revised Text: "Although EBCS are not present in all reconnection events, they emerge robustly during prolonged antiparallel reconnection exhausts, particularly in plasmas with **moderate beta ($0.1<\beta \lesssim 1$)**."

Current Text: "We show that EBCS are not an artifact of geometry or boundary conditions but instead emerge robustly during quasi-steady reconnection, particularly in plasmas with moderate to high β..."

Revised Text: "We show that EBCS are not an artifact of geometry or boundary conditions but instead emerge robustly during prolonged reconnection, particularly in plasmas with **moderate β)**..."

Current Text: "where β is the upstream plasma beta. This relation indicates that stronger ion heating and acceleration are expected in low-β reconnection."

Revised Text: "where β is the upstream plasma beta. This relation indicates that stronger ion heating and acceleration, and deeper potential wells, are expected at lower β. Furthermore, this also reveals an upper limit for EBCS formation: in high-β regimes ($\beta \gg 1$), the potential well becomes too shallow to have distinct EBCS formation. Conversely, in low-β regimes, while the well is deep, the current sheet is highly susceptible to fragmentation via secondary instabilities (like LHDI and plasmoids). Thus, coherent EBCS are optimally formed and maintained in an intermediate, moderate-β regime ($0.1<\beta \lesssim 1$), consistent with the conditions of our three analyzed events.

3. *To establish how the formation of EBCSs depends on β, they perform a simulation with smaller β (Fig. 8). In Fig. 8, they show that although the bifurcated current sheets are formed, they are not extended. Equation (5), or any other equation does not talk about the extension of the EBCSs. It would be good to have a physical and/or analytical understanding about the dependence of the extension of EBCSs on β. Along the same lines, the word "Universal" in the title should be changed or omitted because obviously there are plenty of regions in reconnection parameter space where EBCSs do not form and it is too strong of a word.*

   We agree with the reviewer on both points. First, the reviewer is correct that the word 'Universal' is too strong given that EBCS formation is bounded by the plasma *β* parameter space. We have removed 'Universal' from the title of the revised manuscript.

   Second, we appreciate the push to clarify the physical dependence of the EBCS **extension** on *β*. While Equation 5 explains the **formation** of the current peaks (by dictating the potential well depth), the downstream **extension** of these sheets depends entirely on the macroscopic stability of the reconnection X-line and the current layer over

time. From our multiple simulation scans, we find that when $\beta < 0.1$, the X-line location does not remain stable long enough to generate an extended, coherent EBCS downstream due to frequent large plasmoid formation. As a result, the EBCS is truncated and fragmented.

Conversely, while the exact upper limit is complex, Equation 5 suggests that around $\beta \sim 10$, the in-plane electric field becomes too weak to form distinct EBCS. Therefore, the long spatial extension of EBCSs requires a moderate-β regime where the well is deep enough to form the boundary, but the plasma is stable enough to maintain a prolonged, quasi-steady X-line. We have added this physical explanation to the discussion of Figure 8.

The title has been revised to '**Cross-Scale Observations of Expanding Bifurcated Current Sheets in Fast Magnetic Reconnection**'

We have added two sentences in the paragraph describing Fig. 8 in Methods: "Because the downstream extension of an EBCS relies on the macroscopic stability of a quasi-steady X-line, this instability-driven fragmentation causes the initial bifurcated current sheets associated with the X-line at $z = 0$ to disappear beyond $z \approx 7\di$."

"Together with Eq. \ref{eq:phi}, which indicates that high-β regimes suppress the driving electric field, this low-β result illustrates that the coherent spatial extension of EBCSs fundamentally depends on β."

4. *One of the main points of this work is that these structures (EBCSs) explain the Petschek reconnection or fast reconnection without invoking shocks. Nevertheless, they never provide any reconnection rate or reconnection electric field for any system that they have studied here. To elaborate this point, let us take two of their simulations in Figs. 6 and 8. The upstream β in these two cases are 1.2 and 0.05, respectively. Although they find EBCS in Fig. 6, the same is not observed in Fig. 8. Hence, it would be good to have the corresponding reconnection rates, at least, for these two cases to establish that EBCSs can explain fast reconnection.*

   We thank the reviewer for this point. Providing the reconnection rates is indeed crucial for context, and we agree that it helps clarify the relationship between EBCSs, Petschek geometry, and fast reconnection.

   To address the reviewer's point, all the systems studied here exhibit fast magnetic reconnection. The normalized reconnection rate for the quasi-steady MRX experiment is 0.15. For both MMS and PSP events, due to difficulty in determining the reconnection electric field, the reconnection rate cannot be convincingly measured. For the first VPIC simulation (Fig. 2), the rate is R ~ 0.1. Importantly, in the low-\beta VPIC simulation (Fig. 8), the instantaneous reconnection rate varies over time due to continuous plasmoid formation and ejection. However, when averaged over the ion cyclotron time, the

normalized reconnection rate remains fast at R ~ 0.1.

We have added these reconnection rates to the manuscript to explicitly establish this context.

“The normalized reconnection rate for this quasi-steady MRX regime is fast, at $R \approx 0.15$."
“The normalized reconnection rate for this simulation is $R \approx 0.1$.”
“Despite the lack of an extended EBCS, the overall reconnection process remains fast. While continuous plasmoid formation and ejection cause the instantaneous reconnection rate to fluctuate, the time-averaged normalized reconnection rate (over an ion cyclotron period) is $R \approx 0.1$.”

Minor comments:

1. *In Figs. 2a and 3a, they show the variation of out-of-plane current density Jy. However, in Fig. 6a and 8a, they distinctively write out-of-plane “electron” current density Jey. This would imply that the ion current density is relatively negligible which is understandable at sub-ion scales, but this is not explicitly mentioned in the text.*

   We thank the reviewer for pointing out this distinction. The reviewer is correct that at these sub-ion scales, the out-of-plane current is overwhelmingly dominated by the electrons, making the ion contribution negligible ($J_y \sim J_{ey}$).

   We have revised the caption for Fig. 6a:
   Out-of-plane electron current density ($J_{\text{e}y}$), **which is similar to the total current density J_y**, showing a bifurcated structure near the separatrix.

2. *Here, it is mentioned that Fig. 2d displays the plasma density, and how the in-plane electric field makes the ion density increased in the exhaust. However,in Fig. 2d caption ([166]) it is addressed as electron density (ne) for profile. Just a little more clarification would be nice.*

   We thank the reviewer for pointing out this discrepancy in terminology. Because the plasma is quasineutral, the electron density and ion density are essentially identical at these scales ($n_e \approx n_i$). Therefore, the plotted electron density accurately reflects the overall plasma and ion density behavior.

   We have revised the figure caption:
   Electron density (n_{e}) profile, **which represents the ion density also due to quasineutrality in the plasma**, showing enhancement in the exhaust region.

3. *The authors have mentioned that the red curve in Fig. 3b correspondence to fitting a double Harris sheet. i.e., sum of two tanh functions. Since, Bz has a lot of different slopes in different regime,(in Fig. 3b), it would be nice to have corresponding amplitudes*

and half widths of the functions that result in the Bz plotted here.

We thank the reviewer for this suggestion. Providing the exact fitting parameters for the double Harris sheet clearly demonstrates the different spatial scales of the bifurcated structure. The fit is based on two primary $\tanh$ profiles. For the left current layer, the fitted amplitude is 49 Gauss with a half-width of $0.29d_i$. For the right current layer, the fitted amplitude is 57 Gauss with a broader half-width of $0.70d_i$. We have added these specific amplitudes and half-widths to the figure caption.

“The left layer is characterized by the profile $49\tanh[(x + 1.4 \di)/0.29 \di]$ Gauss, while the right layer is fitted to $57\tanh[(x - 1.0 \di)/0.70 \di]$ Gauss."

4. *I do not understand these lines and calculations. I mean how does Ex ~1000 Vm−1 and corresponding E °ø B drift of electrons comes out to be ~ 60 kms−1 near x = −1di?*

   We are grateful to the reviewer for taking the time to verify this calculation. The reviewer's math is entirely correct based on the text we provided, and they have caught an imprecise coordinate reference and a confusing logical flow on our part.

   In the original text, we loosely referred to the location as x ~ -1 d_i. However, as shown in Fig. 3c, the actual peak of the out-of-plane current density on the low-density side is located further out at x ~ -1.4 d_i. We agree that the best way to demonstrate this relationship is to calculate the required electric field directly from the measured local parameters at this exact peak.

   At x ~ -1.4 d_i, the measured current density is J_y ~ 0.5 MA/m^2 and the local electron density is much lower, n_e ~5 x 10^{13} cm^{-3}. Using these values, the required electron drift velocity is ~60 km/s. Given the measured local magnetic field of B ~ 130 Gauss at this location, the required cross-field electric field to drive this drift is ~800 V/m, close to 1000 V/m. To be more specific, the measured E_x at x = $-1.4d_i$ is 900 V/m (18 V change over about 2 cm). We have revised the manuscript to be more specific to the location of the current sheet on the low-density.

   “The inferred in-plane electric field ($E_x \sim 900$ V/m) is sufficient to drive $\mathbf{E} \times \mathbf{B}$ drifts comparable to the measured electron velocities (~ 60 km/s near $x \approx -1.4\di$).”

5. *line [480]: In simulation (Fig. 2d), experiment (Fig. 3e), earth’s magnetotail (Fig. 4b), the electron’s density has been plotted. However, for heliospheric current sheet, the proton’s density is presented. It would be good to know if there is any particular reason.*

   We thank the reviewer for the observation. For the Parker Solar Probe (PSP) event, we plotted the proton density because solar wind electron measurements are frequently contaminated by spacecraft photoelectrons induced by intense solar UV radiation.

Therefore, the proton density serves as a much more reliable proxy for the total plasma density. We have added a brief parenthetical note to the text to clarify this standard observational choice.

Current Text: "Figure 5b shows the proton density..."

Revised Text: "Figure 5b shows the proton density (n_{p}, used here rather than electron density, as solar wind electron measurements are often subject to photoelectron contamination),...

6. line [531-532]: The authors have neglected the pressure term in equation (1), by saying that it contributes only ~10% to the in-plane electric field. At least any physical intuition and/or reference would be nice.

   We thank the reviewer for requesting this physical intuition. We neglected the electron pressure gradient term because the electric field across the EBCS is usually dominated by the V x B term, which represents the magnetic pressure chance. Keeping this term will change Eqn. (4) as:

   $$e\phi_0 \sim (1/n_{av})(\Delta B^2/2\mu_0 - \Delta p_e).$$

   The electron pressure term obviously represents the change of the electron pressure. Physically, while the magnetic pressure drops significantly across the boundary, the electron temperature does not increase much. Instead, the energy from the deep potential well predominantly drives strong ion heating and acceleration [Drake *et al* 2009]. Consequently, the change in magnetic pressure exceeds the change in electron pressure (mostly due to the density increase), making the electron diamagnetic contribution small (~10 %). We have added this physical justification, along with references, to the text to clarify this assumption.

   “Physically, the $\mathbf{V}_{\text{e}}\times\mathbf{B}$ term represents the change in the magnetic pressure across the boundary, while the pressure gradient term represents that of the electron pressure. Because the energy from the deep potential well predominantly drives strong ion heating and acceleration \citep{Drake2009}, the electron temperature does not increase significantly \citep{Yoo2013}. Consequently, the change in magnetic pressure largely exceeds the change in electron pressure, making the electron diamagnetic contribution small.”

7. *line [545-547]: Equation (4) should have a ‘-’ve sign. Also, nav is not defined.*

   We thank the reviewer for the comment. As noted in our response to Major Comment 1, we have explicitly added the definition and physical justification for n_{av} to the manuscript.

   Regarding the sign of Equation 4: we believe the reviewer might be considering the formal electrostatic integration. However, in our formulation, Φ_0 is explicitly defined

as the **depth** of the potential well. Because both the magnetic pressure and the electrostatic potential are higher outside the exhaust boundary and drop toward the center, the difference between the outside and inside yields a positive scalar magnitude for the well depth. Therefore, the positive sign in Equation 4 correctly represents this physical magnitude. To prevent any future confusion, we have ensured the text explicitly refers to Φ_0 as a positive magnitude/depth.

8. *line [572-573]: Why is the velocity required for converting Ex to potential?*

   We thank the reviewer for asking for this clarification. Because the MMS spacecraft records observations as a time series, the electric field is measured as $E_x(t)$. However, calculating the potential well requires a spatial integration ($\Phi = -\int E_x\, dx$). Therefore, the boundary normal velocity (v_n) is required to convert the measured time interval into a spatial distance ($dx = v_n\, dt$). We have slightly expanded this sentence in the text to make this standard time-to-space conversion explicit.

   For MMS, the **time-series** E_x profile is converted into **a spatial** potential using an estimated normal velocity of 40 km/s.

9. *line [789-792]: The equilibrium field configuration in equations (7) and (8) is not correctly written. For example, the argument of tanh should have x, not z. Please check them thoroughly. Lpert is not described anywhere in the text.*

   We thank the reviewer for their careful reading and for catching this error. The reviewer is correct. In our simulations, the normal direction is x and the outflow direction is x. Therefore, the reconnecting background field (B_z) must depend on the normal coordinate (x). The spatial variables x and z in the initialization equations were accidentally swapped during transcription, and we have corrected the equations so that the tanh profile correctly depends on x.
   Regarding L_{pert}, this represented the length scale of the initial magnetic perturbation along the outflow direction. Because we set this equal to the domain length to seed a single primary X-line, we have simply replaced L_{pert} with L_z in the equations to simplify the notation and avoid unnecessary extra variables. We have also explicitly defined L_x and L_z in the revised text.

   "Here L is the half-width of the initial current sheet, and δB_x and δB_z represent the amplitudes of the initial magnetic perturbation. The parameters L_x and L_z are the total dimensions of the simulation domain in the normal and outflow directions, respectively."

10. line [866-869]: Please check the simulation domain to mimic the MRX experiment in Fig. 9. It is mentioned that Lr = 30di and Lz = 60di. However, in Fig. 9a, the domain sizes are quite smaller. Nevertheless, the separation of flux cores of 18di matches well with Fig. 9a.

We thank the reviewer for their careful attention to the simulation details. The reviewer is correct that the plotted area in Figure 9a is smaller than the stated domain size. The dimensions $L_r = 30\ d_i$ and $L_z = 60\ d_i$ represent the full computational domain, which was chosen to encompass the entire MRX physical vacuum chamber to ensure realistic global boundary conditions. Figure 9a displays a zoomed-in view of the central reconnection region between the flux cores to highlight the local dynamics. We have added a brief note to the text to clarify that the figure shows a cropped region of the full simulated chamber.

Current Text: "The simulation domain is a conducting box of size $L_r = 30\di$ and $L_z = 60\di$, discretized into $n_r = 200$ and $n_z = 448$ cells."

Revised Text: "The simulation domain is a conducting box of size $L_r = 30\di$ and $L_z = 60\di$ (designed to encompass the entire MRX vacuum chamber), discretized into $n_r = 200$ and $n_z = 448$ cells. Note that Figure 9 displays a zoomed-in view of the central reconnection region rather than the full computational domain."

Reviewer 4

We sincerely thank the reviewer for their insightful and encouraging evaluation of our work. We are particularly gratified that the reviewer recognizes the significance of EBCS as a kinetic-scale alternative to the classical Petschek shock model. We agree that this framework provides a vital link between two-fluid plasma physics and the large-scale geometry of fast reconnection. In the following sections, we address the reviewer's specific technical comments regarding simulation boundary effects, data processing in MRX, and the interpretation of the MMS and PSP observations. We believe these clarifications have significantly improved the clarity and rigor of the manuscript. See the following response to each concern:

1. [B] Validity The validity of each data interpretation will be discussed in the following sections, but the results are generally considered to be valid, except for Fig 6. A significant change (sudden bending of the separatrices) is observed in z>95di of Fig. 6a, which is unlikely to be continuous with the 20di<z<80di region. If the distance z~100di has any physical meaning, please explain it. Also, please make sure it is not a problem of the simulation technique (e.g., computational domain).

   We thank the reviewer for their careful observation of the simulation results in Figure 6. The reviewer is correct that the sudden bending of the separatrices near $z \approx 95 d_i$ is not a physical feature of the reconnection exhaust. As specified in the manuscript, the simulation domain extends to $z = 100 d_i$ (half-width of the $200 d_i$ total domain). The features observed very close to this edge are numerical artifacts resulting from the open-boundary conditions used to allow plasma and flux to exit the domain.

These boundary effects do not impact the physical interpretation of the EBCS structure, which remains steady and well-defined throughout the vast majority of the exhaust (e.g., the $z < 80 d_i$ region where our 1D analysis is performed). To avoid any confusion for readers, we have limited the plotted range of the horizontal axis to $95d_i$ in the revised Figure 6, cleanly removing the region affected by these numerical boundary effects.

2. [D1] On line 273, the authors state that there are steep gradients of Bz at x≈±2di in Fig. 3b, but this is not accurate. At least the gradient at x=2di is not steep.

   We thank the reviewer for this careful check. We agree that describing the steep gradients as being at $x \approx \pm 2 d_i$ was inaccurate. The steepest gradients in the B_z profile, which correspond to the locations of the bifurcated current sheets, are actually located at $x \approx -1.4 d_i$ (on the low-density side) and $x \approx 1 d_i$ (on the high-density side). These coordinates match the peaks in the derived J_y profile shown in Fig. 3c. We have corrected the text to reflect these specific locations.

   Current Text: "The steep gradients at $x \approx \pm 2d_i$ and the flattened profile in the central region confirm the presence of EBCS."
   Revised Text: "The steep gradients at $x \approx -1.4 d_i$ and $x \approx 1 d_i$, along with the flattened profile in the central region, confirm the presence of EBCS."

3. [D2] On line 253, it is stated that Jy in the MRX experiment was calculated from the gradient of Bz, but Fig. 3c does not appear to be the x-derivative of Bz in Fig. 3b. Even in flattened regions (x<-2di, -1di<x<0.5di, 2di<x), the gradient of Bz in Fig. 3b is greater than 10% of that in the steep region and is never nearly zero as in Fig. 3c. Furthermore, positive Jy value at x≈-3.5di in Fig. 3c is numerically impossible. Please check the data and processing procedures. Furthermore, MRX results (Fig. 3a) are not very uniform in the z direction at z≈6di. How much does dBx/dz contribute to Jy? Will this result in any significant changes to the interpretation?

   We thank the reviewer for this insightful observation regarding the J_y profile. The reviewer is correct that the J_y shown in Fig. 3c does not match a simple derivative of the raw B_z data points in Fig. 3b.

   The reason is that the total measured magnetic field B_z in MRX is a superposition of the plasma-produced field and the vacuum field generated by the external poloidal-field (PF) and equilibrium-field (EF) coils. These external coils contribute a linear gradient to the B_z profile that does not correspond to a local plasma current density. Following the standard analysis procedure for MRX (e.g., Eq. 19 in Yamada et al. 2000), we fit the measured B_z with a composite function consisting of a double-Harris sheet (the tanh components) plus a linear background term.

To isolate the actual plasma current density J_y associated with the EBCS, we only take the spatial derivative of the tanh components of the fit, intentionally omitting the derivative of the linear vacuum term. This ensures that Fig. 3c accurately represents the bifurcated plasma current sheet rather than the total magnetic gradient. The small positive J_y value mentioned by the reviewer at the far edge ($x \sim -3.5d_i$) is a minor numerical artifact of the fitting and subtraction process and does not represent a physical current. We have clarified this in the revised manuscript and added the reference to Yamada et al. (2000).

Current Text: "...The steep gradients at $x \approx \pm 2d_i$ and the flattened profile in the central region confirm the presence of EBCS. From this fitted profile, the out-of-plane current density J_y is derived as shown in Fig. 3c."

Revised Text: "...The steep gradients at $x \approx -1.4\di$ and $x \approx 1\di$ along with the flattened profile in the central region confirm the presence of EBCS. To obtain the plasma current density J_y, the measured B_z is fitted with a profile consisting of a double-Harris sheet and a linear background term. This linear term represents the vacuum magnetic field from the external coils, which carries a gradient but no local plasma current \citep{Yamada2000}. By taking the derivative of only the $\tanh$ components of the fit, we obtain the plasma current density shown in Fig. \ref{fig:mrx}c.."

4. [D3] On line 394, the authors state that significant Ex and Jy were observed simultaneously, as indicated by the three gray-box regions. The first and last Jy events are thought to be responsible for the step-like changes observed in Bz (lines 351 and 368). However, why does the middle Jy event around 09:49:40 not cause any noticeable change in Bz?

   We thank the reviewer for this sharp observation regarding the middle J_y peak in the MMS data. The reviewer is correct that while the first and last peaks coincide with clear step-like changes in B_z (marking 1D separatrix crossings), the middle peak does not.

   We attribute this discrepancy to the **breakdown of the 1D current sheet approximation** due to plasma instabilities. While the EBCS is primarily a 1D structure along the separatrices, instabilities such as the lower hybrid drift instability (LHDI) can cause the current sheet to bifurcate further or fragment into complex 2D or 3D structures. In such multidimensional configurations, the simple 1D Ampère's law relationship between the out-of-plane current and the magnetic gradient no longer holds. Consequently, the spacecraft can measure a significant local current enhancement without recording the macroscopic magnetic shear signature typically associated with a global 1D boundary crossing. We have added this physical clarification to the manuscript.

   Current Text: "...intervals of strong E_x (gray boxes in panels d and e) coincide with enhanced J_{ey}, suggesting that the out-of-plane current is primarily driven by the $E \times B$ drift."

Revised Text: "...intervals of strong E_x (gray boxes in panels d and e) coincide with enhanced J_{ey}, suggesting that the out-of-plane current is primarily driven by the $E \times B$ drift. Notably, the middle J_{ey} peak does not produce a step-like change in B_z. This likely indicates a transition from a quasi-1D current sheet into a complex 2D or 3D structure driven by instabilities like LHDI. In such multidimensional regions, the 1D relationship between magnetic gradients and out-of-plane current is locally violated, allowing for significant current density without the macroscopic B_z signatures observed at the primary separatrix crossings

5. [D4] Potential in MRX Regarding the potential profile shown in Fig. 3d, the authors state on line 285 that it is "centered at x=0 with sharp drops near x≈±2di", but the plot does not look like that. Also, is there a difference in the potential of both upstream regions? If so, what determines this?

   We thank the reviewer for their careful visual inspection of the data. We fully agree that our original description ("centered at $x=0$ with sharp drops near $x \approx \pm 2d_i$") was inaccurate and painted an overly symmetric picture of the profile.

   In reality, because the plasma is highly asymmetric, the minimum of the potential well is shifted toward the low-density side. Furthermore, the gradient is extremely sharp on the low-density side but much more gradual on the high-density side. We have corrected the text to accurately describe these coordinates.

   Regarding the difference in the upstream potentials: yes, there is a clear baseline potential difference of approximately 13 V between the low-density upstream and the high-density upstream. This macroscopic potential difference is determined by the global density asymmetry across the current sheet; the strong electron pressure gradient establishes a global ambipolar electric field that offsets the baseline plasma potentials. This behavior is consistent with our previous studies of asymmetric reconnection in MRX (e.g., Yoo et al., 2014). We have updated the manuscript to explicitly note and physically explain this upstream difference.

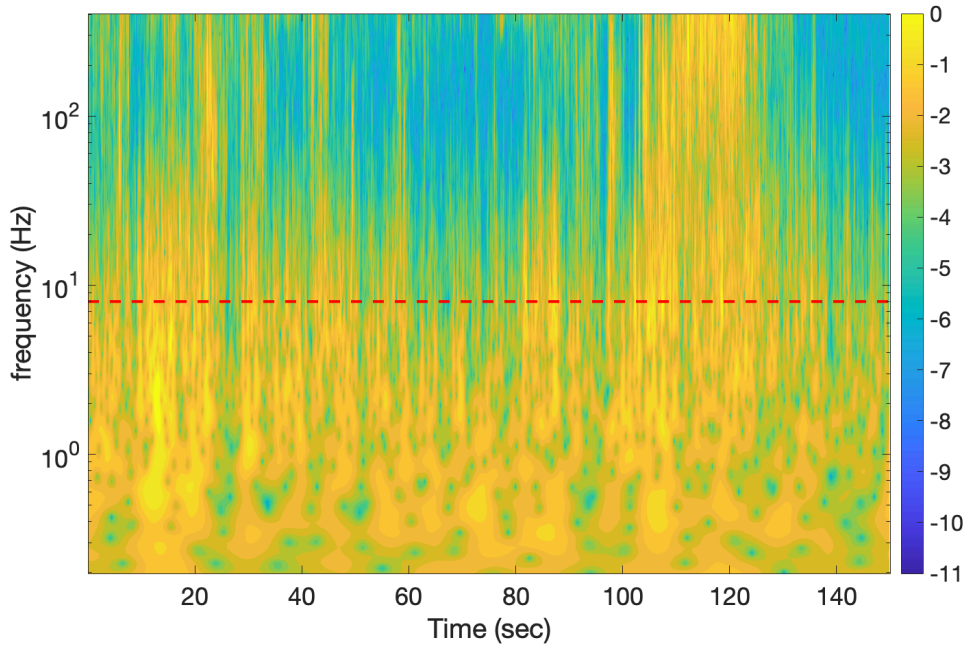
   Current Text: "The resulting profile reveals a distinct potential well centered at $x=0$ with sharp drops near $x \approx \pm 2d_i$. The well is asymmetric, with a steeper gradient on the low-density side, consistent with previous MRX results \citep{Yoo2014b}."

   Revised Text: "The resulting profile reveals a distinct, asymmetric potential well whose minimum is shifted toward the low-density side ($x \approx -0.8 d_i$). The potential exhibits a sharp drop near $x \approx -1.5 d_i$ on the low-density side, whereas the change on the high-density side is much more gradual. Furthermore, there is a noticeable baseline difference of ~ 13 V between the two upstream regions; this offset is determined by the global density asymmetry, which drives a macroscopic ambipolar electric field across the entire current sheet layer \citep{Yoo2014b}."

6. [D5] Ex in MMS It is stated that the Ex fluctuations observed by MMS shown in Fig. 4d are due to LHDI (line 511). If possible, please provide the results of frequency analysis.

We thank the reviewer for this excellent suggestion, which strengthens our identification of the instability. To confirm the presence of LHDI, we performed a wavelet frequency analysis of the E_x fluctuations observed by MMS during the 150-second interval encompassing the first separatrix crossing (09:43:53 to 09:46:23 UT, just before the first crossing of EBCS).

As shown in the spectrogram provided below for your reference, the red dashed line denotes the local lower hybrid frequency (~8 Hz, calculated using the local magnetic field of ~12 nT). The spectrum reveals that the fluctuation energy is broadband and predominantly concentrated below f_{LH}. This spectral morphology is a classic, established signature of LHDI operating at reconnection boundaries [Carter et al. 2002, Roytershteyn et al. 2012]. To maintain visual clarity in Figure 4, we have opted not to add a spectrogram panel to the main text. Instead, we have added a sentence to the manuscript explicitly summarizing the results of this frequency analysis to quantitatively support our physical interpretation.



Current Text: "The fluctuations in E_x shown in Fig. \ref{fig:mms}d near the two separatrices are associated with this instability."
Revised Text: "The fluctuations in E_x shown in Fig. \ref{fig:mms}d near the two separatrices are associated with this instability; wavelet analysis of E_x reveals broadband fluctuation energy concentrated predominantly just below the local lower hybrid frequency ($f_{LH} \approx 8$ Hz), which is a characteristic signature of LHDI

\citep{Carter2001,Roytershteyn2012}."

7. [F] While I do not believe that additional experiments are necessary, it would be very powerful to show data supporting their claim that the formation of bifurcated current sheets strongly depends on plasma β (line 717). References 12 and 17 conduct statistical analyses of the presence or absence of current sheet bifurcations, but how consistent are they with the authors' opinion?

   We thank the reviewer for pointing us to References 12 and 17. We fully agree that connecting our findings to these statistical studies strongly bolsters our claim. Asano et al. (Ref 17) found that bifurcated current sheets are 'atypical' in the near-Earth magnetotail (a lower-β, highly bursty environment), whereas Gosling & Szabo (Ref 12) observed they are common in the heliospheric current sheet (a higher-β, quasi-steady environment). This statistical space data aligns perfectly with our physical framework and our low-β simulation results.

   Regarding a statistical analysis of the β dependence in our laboratory data, this is unfortunately difficult to achieve in MRX due to its limited accessible parameter space and natural shot-to-shot variation. The 250 discharges used in our analysis were explicitly selected because they exhibited clear EBCS signatures, meaning they all shared similar β and stable conditions. We found that MRX discharges lacking EBCS signatures typically suffered from unstable X-lines or extreme asymmetry. This strongly supports our conclusion that 'quasi-steady' dynamics are strictly required alongside a moderate β. A comprehensive laboratory statistical study exploring a wide β parameter space will be a target for next-generation reconnection facilities, such as the FLARE device.

   We have updated the discussion paragraph in the revised manuscript to explicitly integrate the statistical context from Refs 12 and 17, and we have added a note regarding the experimental constraints and future laboratory investigations.

   The infrequent observation of EBCS in the near-Earth magnetotail can be attributed to the rapidly changing global boundary conditions of the magnetosphere, which rarely allow for prolonged quasi-steady flux annihilation. This scarcity aligns precisely with statistical surveys characterizing bifurcated current sheets as relatively rare or ``atypical'' in the magnetotail \citep{Asano2005}. In contrast, the global conditions of the heliospheric current sheet evolve much more slowly, allowing macroscopic boundary structures like those observed by PSP to persist, a physical reality supported by the statistically frequent observation of bifurcated sheets in the solar wind \citep{Gosling2008}. Furthermore, this steadiness is strongly mediated by the plasma β. In moderate β environments (such as MRX and the PSP event), global quasi-steadiness is maintained. While exploring a comprehensive range of β in MRX is challenging due to the limited accessible parameter space, making this an objective for next-generation facilities such as FLARE (Facility for Laboratory

Reconnection Experiments), we note that MRX discharges lacking EBCS signatures typically has unstable X-lines, reinforcing the strict necessity of quasi-steady dynamics. In low-β environments typical of the magnetotail, the current sheet associated with EBCS is highly susceptible to the lower hybrid drift instability (LHDI) \citep{Le2018}. In addition, large-scale plasmoids are formed, making the reconnection globally bursty (Fig. \ref{fig:lowbeta}), thereby preventing the formation or persistence of coherent EBCS. When global quasi-steadiness is achieved, however, EBCS serve as robust boundary structures regardless of local bursty dynamics at the X-line."

8. Clarity and context What does "Expanding" in the title mean? Does it mean constant expansion over time? The data and text of this paper do not show how the structure evolves over time, so it's difficult to imagine what it might look like. Please explain why the authors named it "Expanding."

   Regarding "strong in-plane electric field along separatrices" on line 120, I believe the spatial distribution of E is "along" but the direction of E is "across." Please write in a way that avoids misunderstandings.

   On line 714, "Simulations in Extended Fig. 3 supports this view" - does this refer to Fig. 8?

   We thank the reviewer for pointing out these areas where the text needed clarification.

   Meaning of 'Expanding': A similar point was raised by Reviewer 1. To clarify, 'Expanding' in the title refers to the **spatial divergence** of the bifurcated current layers as they extend downstream from the X-line, forming the open, Petschek-like exhaust. It does not imply a temporal expansion or transient growth over time. Because the reconnection is quasi-steady, this spatially expanding structure is stable in time. We have clarified this spatial meaning early in the revised manuscript.
   “These structures are termed `expanding' because the distance between the double-peaked current layers naturally increases downstream as they follow the diverging separatrices.”

   Direction of the Electric Field: The reviewer is correct. The spatial *distribution* of the electric field is along the separatrices, but the *direction* of the electric field vector is across (normal to) them. We have corrected the text:

   “This potential structure creates a deep potential well that spans the entire macroscopic exhaust region. However, the steep potential drops (and therefore the strongest in-plane electric fields, primarily oriented along the inflow direction, E_x) are highly localized to the narrow separatrix regions bounding the exhaust.”

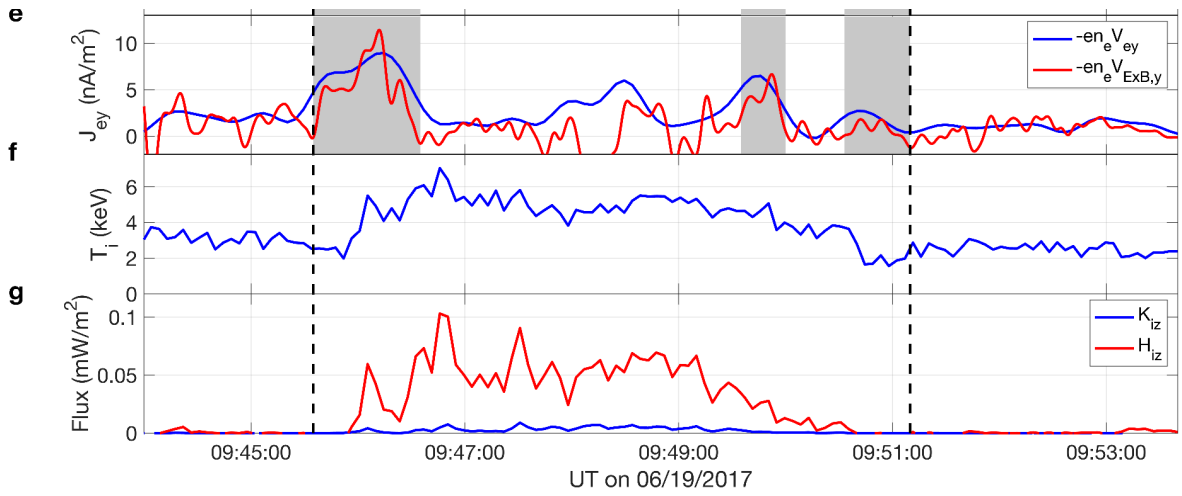
   Figure Reference Typo: Yes, the reviewer is correct. 'Extended Fig. 3' was a typographical error from an earlier draft format and was meant to refer to **Figure 8**. We have corrected this reference in the text.

Response to Reviewers' Comments

Reviewer 1

We thank the reviewer for valuable comments. We have addressed the comments in the revised manuscript by revising two figures and including some details. Our detailed point-by-point responses and the corresponding manuscript revisions are provided below.

1. In the MMS event, the authors claim that the current layer Jey is generated by the E×B drift. Since the full 3D electric and magnetic field vectors are available from MMS measurements, it should be possible to directly verify whether Jey can indeed be explained by Nq(E×B) or not. My concern arises because previous magnetotail observations have shown that the current along the separatrices is often dominated by the parallel component, for example, as shown in Fig. 2e of DOI:10.1002/grl.50537. Moreover, just before the current sheet crossing discussed in this paper, there is another separatrix crossing event in which the current is also dominated by the parallel component (https://doi.org/10.1029/2019GL082538).



We thank the reviewer for this insightful comment and for pointing out the relevant literature regarding parallel currents along separatrices in the magnetotail. Prompted by your suggestion, we have updated our analysis to directly verify the source of the current by calculating the perpendicular ExB drift current density. We have added this calculated drift current (red line) alongside the measured electron current J_{ey} = -en$_e$ V_{ey} (blue line) in a newly revised panel e. As shown above, there is excellent agreement between the measured J_{ey} and the ExB drift current specifically near the separatrix regions (the gray shaded areas). This directly confirms our claim that the current layer at these boundaries is indeed dominated by the perpendicular ExB drift. Figure 4 has been replaced with this new figure.

Revised: “Figure \ref{fig:mms}e displays the out-of-plane electron current density (blue line, $J_{\text{e}y}$), which is low-pass filtered to remove high-frequency noise. To verify the physical origin of this current, we also plot the out-of-plane current density expected purely from the local $\mathbf{E} \times \mathbf{B}$ drift (red line, $-en_e V_{E \times B, y}$). The results show that $J_{\text{e}y}$ is not concentrated at the exhaust center, but instead enhanced near the separatrices, consistent with the EBCS structure. Moreover, intervals of strong E_x (gray boxes in panels d and e) perfectly coincide with these current enhancements. The excellent agreement between the measured electron current and the calculated drift current in these bounded regions confirms that the out-of-plane electron current is indeed primarily driven by the $\mathbf{E} \times \mathbf{B}$ drift within the electrostatic potential well.

2. Lines 290–296: How was the 2D current profile shown in Fig. 3a obtained in the laboratory experiment? From the manuscript, it appears that the current density shown in Fig. 3c was derived from the magnetic field Bz, but the procedure for obtaining the 2D current profile is not sufficiently clear.

   Unlike Fig. 3c, the 2D profile is not derived from a simple 1D extrapolation. Instead, it is calculated from direct 2D measurements of the full magnetic field vector. We utilize a 2D array of internal magnetic probes that simultaneously measure all three magnetic field components across a 2D spatial grid consisting of more than 100 measurement points. The out-of-plane current density is then calculated directly from the measured in-plane magnetic field components (B_x and B_z) using Ampere's law: $J_y = (\partial B_x/\partial z - \partial B_z/\partial x)/\mu_0$. We have revised the text describing Fig. 3a to explicitly detail this experimental procedure and grid resolution.

   "The 2D current profile shown in Fig. \ref{fig:mrx}a is obtained from direct measurements of the full magnetic field vector. Using the 2D magnetic probe array, the magnetic field components are simultaneously measured across a 2D spatial grid. The 2D out-of-plane current density (J_y) is then directly calculated from the measured in-plane magnetic fields (B_x and B_z) using Amp\`ere's law, $J_y = \left(\partial B_x/\partial z - \partial B_z\partial x \right)/\mu_0$."

3. Lines 328–336: It appears that the density dip is observed on the right-hand side at x=2.0, rather than at x=1.6di. Please check this description carefully.

   We thank the reviewer for their careful reading of our manuscript and figures, and for catching this oversight. The reviewer is completely correct; the density dip is indeed centered near $x = 2d_i$ rather than $1.6d_i$. We have corrected this typographical error in the revised text.

4. Lines 500–505: The times 08:14:30 and 08:28:15 do not appear to correspond to the sudden increase and sharp decrease described in the text. I could not identify these features at the indicated times.

We sincerely thank the reviewer for their keen eye in catching this discrepancy. The reviewer was correct that the features did not align with the time axis in the previous draft. Upon investigating, we discovered that the timestamps described in our text (08:14:30 and 08:28:15) were indeed correct, but a plotting error had introduced a 3-minute offset to the horizontal time axis of Figure 5.

We have corrected the time axis in the revised figure. As shown in the updated plot, the sudden increase and sharp decrease (indicated by the dashed vertical lines) now correspond to the times stated in the manuscript text.

Reviewer 2 and 3

We thank the reviewers for their continued rigorous evaluation of our manuscript. We deeply appreciate the push for precise physical definitions and a strictly controlled numerical parameter study. Your critical feedback has been instrumental in strengthening the physical arguments and empirical foundation of this paper. We have carefully addressed all of your concerns, including running a new suite of tightly controlled particle-in-cell (PIC) simulations to explicitly isolate the plasma beta dependence. Our detailed point-by-point responses and the corresponding manuscript revisions are provided below.

Major Comments:

1. *The authors in fact have not even defined what EBCS actually are. In the references they cite [7, 11-15], the EBCS can extend up to hundreds or thousands of di. However, in their Figures 2 (simulation) and 3 (experiment), what they call EBCS only extend up to several di. If several di extension is sufficient for EBCS, then the current sheet bifurcation in Figure 8 also extends from 0-6 di and therefore is an EBCS. But they present Figure 8 as a regime where EBCS does not exist. Therefore, the definition of EBCS should be given clearly as currently it is very ambiguous, and calling a bifurcated current that extends only several di as EBCS is questionable.*

   We thank the reviewers for raising this important point regarding the exact definition and spatial extent of the EBCS. We agree that our initial definition in the introduction could be formulated more precisely to avoid this exact confusion.

   To clarify: the definition of an EBCS does not depend on its absolute length. Physically, an EBCS is defined as a bifurcated current sheet that forms due to localized two-fluid effects and expands outward along the magnetic separatrix. Because this is a fundamental kinetic mechanism, the initial bifurcation occurs near the X-line in most symmetric collisionless reconnection events, including the low-beta case shown in old Figure 8 (where it extends to about 6–7 di). However, the central thesis of our manuscript is that under moderate-beta, quasi-steady conditions, this localized kinetic structure is not destroyed; instead, it persists and extends far downstream, scaling with the system size to form the macroscopic boundaries of a Petschek-like exhaust (e.g.,

extending to the system boundaries in Figures 2 and 3, extending 200 di in Figure 6, and extending 10,000 di in the PSP event).

Conversely, in low-beta regimes, the current sheet is highly susceptible to plasmoid formation and secondary instabilities. As we noted in Lines 971–975 of the previous draft, this instability-driven fragmentation destroys the boundary, causing the initially bifurcated sheets in original Figure 8 to break up and disappear beyond z ≈ 7 di. Therefore, a macroscopic Petschek-like exhaust cannot form in that regime.To eliminate any ambiguity, we have explicitly updated the Introduction to define the fundamental physical structure of an EBCS, highlight the critical distinction between local formation and macroscopic persistence, and state explicitly that in this manuscript, the term 'EBCS' is used exclusively to refer to structures that extend far downstream, comparable to the system size."

Current Text: "EBCS consist of double-peaked current layers that extend along magnetic separatrices, accompanied by sharp magnetic field variations, localized electric fields, and enhanced ion outflows. These signatures bear resemblance to the open exhaust geometry envisioned by Petschek [18, 19], yet the physical mechanisms..."

Revised Text: "Physically, EBCS are defined as double-peaked current layers that are generated by two-fluid effects and expand outward along magnetic separatrices. Because they are rooted in kinetic-scale decoupling, the initial formation of these bifurcated sheets occurs near the X-line in most symmetric collisionless reconnection events. However, their macroscopic spatial extent depends heavily on the global plasma conditions. In bursty, low-β environments, the boundary is quickly destroyed by secondary instabilities and plasmoid fragmentation. In contrast, under quasi-steady conditions, these kinetic boundaries can persist and extend far downstream, scaling with the system size to form the stable, open exhaust geometry originally envisioned by Petschek \citep{Petschek64}. Throughout this manuscript, the term EBCS is used exclusively to refer to these macroscopically extended boundaries that persist far downstream, comparable to the system size.

EBCS are accompanied by sharp magnetic field variations, localized electric fields, and enhanced ion outflows. These signatures bear resemblance to the open exhaust geometry envisioned by Petschek \citep{sato79,UK2000}, yet the physical mechanisms responsible for their formation and their role in reconnection dynamics have not been fully understood \citep{Yamada2022}."

2. *Thanks to the authors' explanations, now we understand what Equation 5 means. Let us first point out two minor points. First, it should be made clear that the electron pressure in Eq. 1 is the local electron pressure but the beta in Equation 5 refers to the global beta. Second, the beta should be properly defined: is it the electron beta, the ion beta, or total (the ratios are all different in the simulations). Now the main point: the expression in Equation 5 tells you nothing about the length scale of the potential drop. This is*

important because the existence of a large potential drop alone does not explain the localized nature of the current sheet and the electric field. The localized current on each side of the bifurcation signifies a localized Hall field, which means the potential drop must happen over a short distance. Why it occurs over such short distance is not addressed by Eq. 5 and its derivation. If the same potential drop occurs over a much larger spatial distance, such localized current sheets would not arise. Therefore, the present analytical derivation does not address the origin of EBCS.

We thank the reviewers for their careful evaluation of our analytical formulation. We have fully addressed the two minor points in the revised text: we now explicitly define p_e in Equation 1 as the local electron pressure, and we clarify that beta in Equation 5 refers to the total global upstream plasma beta.

Regarding the main point, the reviewer is correct that Equation 5 determines the magnitude of the potential drop rather than its cross-field length scale. Explaining this spatial scale and its downstream evolution are critical to the origin of the EBCS.

As discussed in our two-fluid framework, the potential drop originates from charge separation near the X-line, where the cross-field scale is strictly constrained to the narrow electron kinetic scale. Under quasi-steady conditions, highly mobile electrons map this potential well far downstream along the newly reconnected field lines.

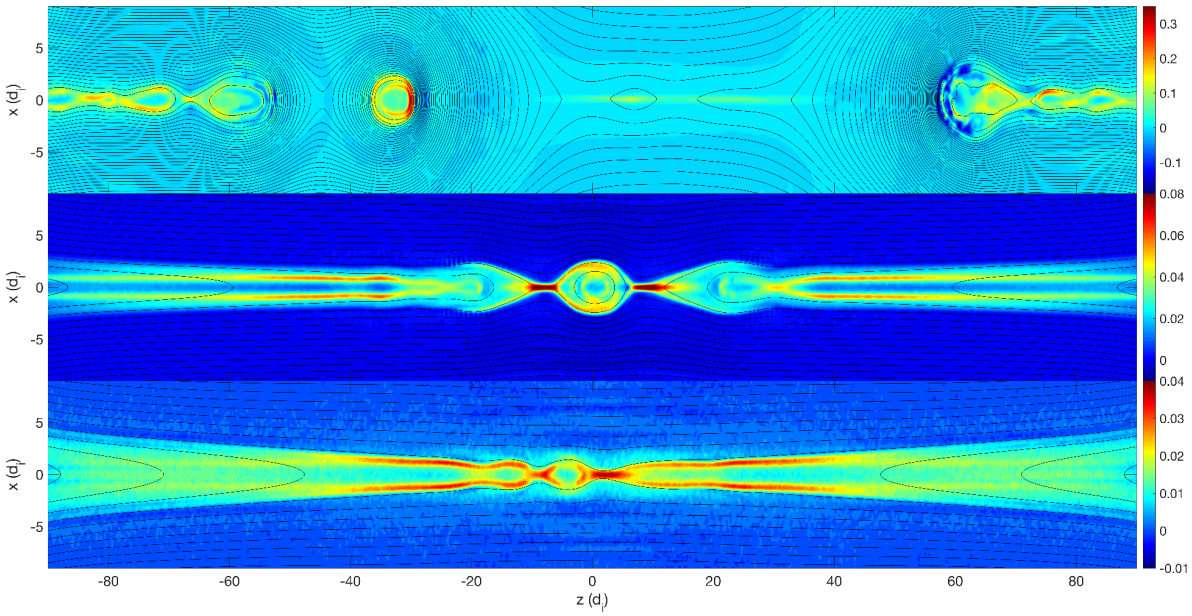
However, as the structure extends macroscopically, the boundary naturally broadens from sub-d_i scales in MRX, to several d_i in MMS, to ~20d_i in PSP. Because the potential profile is dictated by electron dynamics, this gradual broadening is governed by electron finite gyro-radius effects. As electrons deliver the potential information further downstream, their cross-field motion gradually weakens the sharpness of the spatial scale across the magnetic field. This combination of a macroscopic potential magnitude (governed by Eq. 5) mapped along the separatrix, whose cross-field thickness originates at the electron scale and gradually broadens downstream produces the localized electric field driving the EBCS.

Furthermore, this mapping explains why EBCS extension is limited in low-beta regimes: if the current sheet is disrupted by plasmoids or lower hybrid drift instability (LHDI), the coherent transmission of this potential is broken. We have added a comprehensive paragraph immediately following Equation 5 to explicitly connect the potential magnitude to this spatial evolution.

Revised Text (Near Eq. 1): "...where the electron pressure tensor is approximated as scalar, **and p_{e} represents the local electron pressure.**"
Revised Text (Near Eq. 5): "...where β is the **total upstream plasma beta**

While Eqn. \ref{eq:phi} establishes the macroscopic magnitude of the potential well, the cross-field spatial scale of the potential drop evolves dynamically. The charge separation generating this potential originates near the X-line, where the initial thickness is strictly constrained to the narrow electron kinetic scale. Under quasi-steady conditions, highly mobile electrons map this potential structure far downstream along the separatrices. However, as the exhaust extends macroscopically, the boundary naturally broadens from sub-$\di$ scales in MRX, to several $\di$ in MMS, to roughly $20\di$ in the PSP event. Because this potential profile is fundamentally governed by electron dynamics, this spatial broadening is driven by electron finite gyro-radius effects; as electrons deliver the potential information further downstream, their cross-field motion gradually weakens the sharpness of the spatial gradient across the magnetic field. This steep, though evolving, spatial gradient, combined with the large potential magnitude, results in the localized in-plane electric field that drives the $\mathbf{E} \times \mathbf{B}$ current of the EBCS. Conversely, if the current sheet is unstable to secondary instabilities such as plasmoid formation \citep{Daughton2011b} or the lower hybrid drift instability (LHDI) \citep{Le2018}, the coherent transmission of this potential information along the magnetic field is disrupted, thereby fundamentally limiting the downstream extension of EBCS.

3. *If the authors wish to make the argument that the formation of EBCS depends on the plasma beta, they should perform comparisons between systematically varied simulations. Right now, the four PIC simulations, including the one in the Appendix, all have differing domains, mass ratios, temperature ratios, temperature, sheet widths, etc. The length scales in electron skin depths, which are the species that control the EM field (e.g., Eq. 1) are all over the place. There isn't a single set of two simulations with only beta as the different parameter. It is thus impossible to isolate the cause of the EBCS through these simulations. Also, the plasma beta for the simulation in Figure 2 is not even mentioned, and from Reference 27, we only guess it to be 0.03. This is similar to the simulation in Figure 8, which the authors show as an example of absence of EBCS (this point is also related to point 1 above).*



We sincerely thank the reviewers for their rigorous evaluation and for highlighting the need for clarity regarding our simulation parameters.
First, regarding the simulation in Figure 2: The reviewer made a very reasonable deduction based on Reference 27, which indeed yields a value of 0.03. However, this value corresponds strictly to the *electron* beta. Because the simulation utilizes a high ion-to-electron temperature ratio (T_i / T_e = 5), the *total* global upstream plasma beta is actually 0.16. We apologize for not explicitly stating this in the previous draft. Therefore, Figure 2 represents a moderate-β regime, which perfectly aligns with our central claim that EBCS structures stably form and extend in moderate-β environments. We have updated the manuscript to explicitly state the total β = 0.16 for Figure 2.

However, we agree with the reviewer's broader and more critical point: our previous suite of simulations varied multiple parameters simultaneously (domain size, mass ratio, temperature ratio, etc.), making it difficult to definitively isolate plasma β as the sole controlling factor.

To directly address this, we have completely replaced Figure 8 with a new series of strictly controlled 2D PIC simulations. Crucially, the numerical and physical parameters for these new runs (domain size, grid resolution, initialization, and mass/temperature ratios) are kept exactly identical to those used in Figure 6 (which successfully demonstrated EBCS formation at β = 1.2, matching the PSP observation). The *only* parameter varied across this entire suite is the global upstream plasma β.

The new Figure 8 presents the out-of-plane current density for three cases:

- Top Panel (β = 0.012): In this low-β regime, the macroscopic current sheet is highly susceptible to plasmoid instability. Frequent, large-scale plasmoids form

and dominate the dynamics, preventing any coherent EBCS from extending downstream.

- Middle Panel (β = 0.12): At this intermediate value, although frequent plasmoids still form near the central current sheet, distinct bifurcated current sheets (EBCS) are now clearly able to establish themselves and successfully extend into the downstream exhaust region.
- Bottom Panel (β = 0.36): In this moderate-β regime, a clear EBCS forms and extends far into the downstream region (z = 50 d_i).

Unlike the β = 1.2 case, the EBCS in the β = 0.12 and β = 0.36 cases do not extend to the system boundaries. Before the EBCS can fully develop, a plasmoid forms near the center of the current sheet and propagates downstream, significantly disrupting the EBCS structure. This demonstrates that the formation and macroscopic extension of the EBCS require a quasi-steady current sheet. This newly added controlled parameter study explicitly isolates the physical conditions responsible for EBCS formation. It provides definitive evidence that moderate-to-high β plasmas promote the macroscopic extension of the EBCS by suppressing plasmoid fragmentation, thereby maintaining the quasi-steady conditions necessary for its development.

Reviewer 4

We sincerely thank the reviewer for their positive assessment of our revisions and for their encouraging words regarding the enhanced value of the paper. We deeply appreciate your careful reading, which caught this error. You are correct; while we drafted the revised text for Comment [D3] in our previous response, it was inadvertently left out of the main manuscript file during the final document revision. We have revised the manuscript to include the missing content.

In this paper, the authors investigated the reconnection separatrix region in different plasma environments, including magnetosphere, solar wind, and laboratory experiment, via spacecraft observations, laboratory experiment and PIC simulations. They concluded that the expanding bifurcated current sheets (EBCS) are universal structure for quasi-steady reconnection regardless of the system size. The separatrix region is indeed an important region during magnetic reconnection, both in the planet's magnetosphere and solar wind, and shows very dynamical micro-physics. However, the features along separatrix region during reconnection in solar wind and in the magnetosphere are dramatically different. In the magnetosphere, enhanced plasma waves, especially electrostatic waves, strong parallel electric field structure, plasma density peak/cavity, temperature decreases, intense electron beam parallel to the magnetic field, and strong parallel current are generally observed. Of course, the sub-alfvenic ion jets are bounded by the separatrix regions. Some of these features are consistent with EBCS in this paper. On the other hand, in the solar wind, a pair of the slow-mode shocks or discontinuities are detected in the boundary of reconnection-related sub-alfvenic ion jets which has been thought to be quasi-steady reconnection. In this paper, the separatrix region are called EBCS. However, the features along the EBCS in the solar wind are quite different from those in the magnetosphere. For example, in the solar wind, the sharp magnetic field gradient regarded as current layer was detected, without enhanced waves, electron beams etc. So, this reviewer does not think the authors have found many common features along the boundary of the reconnection outflow jets.

Another main issue is the claim of quasi-steady reconnection. In the magnetosphere, the reconnection both at the magnetopause and in the magnetotail are bursty rather than quasi-steady reconnection, even it can persistent for hours. In PIC simulations, the reconnection rate is generally temporal variation and cannot be steady also. In the solar wind, the ion jets or exhausts are generally called quasi-steady reconnection, since the similar flows were repeatedly observed by different spacecrafts during hours (e.g., Phan et al., nature 2006). However, this explanation is not unique. The steady ion jets do NOT mean the corresponding reconnection is steady also. The reconnection could be still bursty and the resulting outflow jets keep steady during their propagation in the interplanetary space. Recently, based on an MMS observation in the solar wind, it shows that the reconnection responsible for the steady outflows is bursty and turbulent (e.g., Wang et al., Nature Astronomy 2023). So, this reviewer suggests avoid the steady reconnection in the paper, because the reconnection in the magnetosphere and laboratory must be bursty.

In my view, this paper mainly studied the separatrix region or EBCS. The separatrix region and the exhaust region are frequently confused throughout the paper. The separatrix region should be the region between the separatrices (the first reconnecting magnetic field line) and the reconnected field line. It is a narrow area in the inflow direction and can extend very far along the outflow direction. The potential well and the ion outflow occupy the whole exhaust region not only the separatrix region. In the paper, these two regions should be distinguished evidently. Otherwise, it is different to know what EBCS exactly refers to.

When comparing the laboratory result, magnetotail observation, and the solar wind observation, in my view, the magnetic field Bz among all three events are quite different. In the solar wind, the Bz gradient at the two vertical dashed lines are very steep. But, the claimed steed Bz gradient in Figs 3 and 4 is much weaker. In the magnetotail (Fig. 4), the current density is not two peaks but at least four peaks, thus the bifurcated current sheet should be inaccurate. The normal electric field Ex in the Fig. 4 was negative in the left half of the current sheet and mainly located in the separatrix region. It seems Ex in the following half was not always positive as claimed in the paper. The electron flow was not shown here. Is it consistent with E_cross_B here? How did the authors get the coordinates in Fig. 4? It is largely different from GSE/GSM coordinates. About the solar wind event, did the authors check the Rankine–Hugoniot relations at the sharp boundary ?

According to the PIC simulations, in Fig. 2 and Fig. 6, one can find clear evolution of current density and plasma velocity along the z direction. It indicates that the structures in separatrix region should

depend on the spatial scale along the outflow direction rather than regardless of the system size claimed in the paper. There are a few other questions shown below. Because these questions, this reviewer cannot recommend this paper at present.

Lines 77-80: actually, the slow-mode shock has been extensively observed in the solar wind and in the magnetotail also. Its relation to the reconnection is indeed unclear.

Line 81: what does the word of “expanding” mean in EBCS? I did not find any discussion on expanding in the paper.

Lines 174-181: The potential well centered at x=0 and across the exhaust region has been verified by spacecraft measurement, consistent with the Hall electric field. It is shown in the paper also. But, the significant potential drops along the separatrices haven’t be shown in this paper. In the exhaust region, the electrons should experience E_cross_B drift. In the sepratrix region, the inflowing electron beams along the magnetic field should be there.

Lines 190-206: Where is the sharp increase of the ion outflow Viz ? Fig. 2c shows the ion flow occurs everywhere in the outflow region, and a further enhancement along the separatrix region between z = (2,4). If this enhancement along the separatrix refers to the Petscheck-type exhaust, it is very localized and depended on the z axis, and not a common feature even along the separatrices.

Lines 400-405: the temperature data were used to calculate the ion enthalpy flux. However, the temperature is inaccurate in the MMS event, because the thermal ions cannot be covered by the FPI instrument here. This introduces a potentially large uncertainty.

In the manuscript "Expanding bifurcated current sheets: universal structure for quasi-steady magnetic reconnection," J. Yoo and others study expanding bifurcated current sheets (EBCS) through simulation, experiment (MRX), and data from spacecraft missions (MMS and PSP). EBCSs are characterized as: double-peaked current layers expanding along the magnetic separatrices, sharp magnetic gradient, localized E-field, strong ion-outflows. These structures appear to be recurring and fundamental at the quasi-steady state of reconnection in plasmas with high β.

These signatures of EBCSs appear due to two fluid effects, that generate electrostatic potential well. They provide a simple estimation of that potential well too.

This model may potentially explain fast magnetic reconnection without invoking the inconclusive slow-shock feature of Patschek model of fast reconnection. The EBCSs, formed naturally due to two-fluid dynamics, have Petschek-like open exhaust. Thus, it provides fast magnetic energy conversion in system of any size.

Although the manuscript is interesting, there are major points that must be addressed adequately through revisions and responses before we can recommend acceptance.

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Our comments are given below.

1 Major comments:

1. The expression for the potential well Φ_0 is given in equation (4) after integrating equation (1) via equations (2) and (3). The integration is performed across the EBCSs. However, the number density, n is considered as average density n_{av} (not described though in the text). The number density changes significantly across the EBCSs as pointed out in most of the figures, in particular Fig. 3e. Why is it feasible to consider an average density across the EBCSs and take it out of the integration?

2. In equation (5), the authors show that the potential is inversely proportional to the upstream β. According to this expression, large β systems have comparatively small potential well and hence the in-plane electric field E_x, which is the main reason for the presence of EBCS due to the $\mathbf{E} \times \mathbf{B}$ drift of electrons. However, they also claim that the EBCSs preferentially occur in higher β plasma systems. This contradiction is crucial to the manuscript's main point and thus should be resolved fully.

3. To establish how the formation of EBCSs depends on β, they perform a simulation with smaller β (Fig. 8). In Fig. 8, they show that although the bifurcated current sheets are formed, they are not extended. Equation (5), or any other equation does not talk about the extension of the EBCSs. It would be good to have a physical and/or analytical understanding about the dependence of the extension of EBCSs on β. Along the same lines, the word "Universal" in the title should be changed or omitted because obviously there are plenty of regions in reconnection parameter space where EBCSs do not form and it is too strong of a word.

4. One of the main points of this work is that these structures (EBCSs) explain the Petschek reconnection or fast reconnection without invoking shocks. Nevertheless, they never provide any reconnection rate or reconnection electric field for any system that they have studied here. To elaborate this point, let us take two of their simulations in Figs. 6 and 8. The upstream β in these two cases are 1.2 and 0.05, respectively. Although they find EBCS in Fig. 6, the same is not observed in Fig. 8. Hence, it would be good to have the corresponding reconnection rates, at least, for these two cases to establish that EBCSs can explain fast reconnection.

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2 Minor comments:

1. In Figs. 2a and 3a, they show the variation of out-of-plane current density J_y. However, in Fig. 6a and 8a, they distinctively write out-of-plane "electron" current density J_{ey}. This would imply that the ion current density is relatively negligible which is understandable at sub-ion scales, but this is not explicitly mentioned in the text.

2. line [200-206]: Here, it is mentioned that Fig. 2d displays the plasma density, and how the in-plane electric field makes the ion density increased in the exhaust. However,in Fig. 2d caption ([166]) it is addressed as electron density (n_e) for profile. Just a little more clarification would be nice.

3. line [272-273]: The authors have mentioned that the red curve in Fig. 3b correspondence to fitting a double Harris sheet. i.e., sum of two tanh functions. Since, B_z has a lot lot of different slopes in different regime,(in Fig. 3b), it would be nice to have corresponding amplitudes and half widths of the functions that result in the B_z plotted here.

4. line [287-289]: I do not understand these lines and calculations. I mean how does $E_x \sim 1000$ Vm^{-1} and corresponding $\mathbf{E} \times \mathbf{B}$ drift of electrons comes out to be ~ 60 kms^{-1} near $x = -1d_i$?

   If we consider $\mathbf{E} \times \mathbf{B}$ drift of electrons to be ~ 60 kms^{-1} $= 60 \times 10^3$ ms^{-1}, then let us calculate the corresponding J_y from the electron density at $x = -d_i$ from Fig. 3(e).

   $n_e \simeq 10 \times 10^{13}$ cm^{-3} $= 10 \times 10^{13} \times 10^6$ m^{-3} $= 10 \times 10^{19}$ m^{-3} at $x = -d_i$.

   Hence, $|J_y| = en_eV_e = 1.6 \times 10^{-19} \times 10 \times 10^{19} \times 60 \times 10^3$ A m^{-2}

   $= 960 \times 10^3$ A m^{-2} $= 0.9$ MA m^{-2}.

   However, from Figs. 3a and/or 3c, $|J_y| \simeq 0.2$, which is around 4 to 5 times smaller than the inferred value of $|J_y|$ at $x = -d_i$.

5. line [480]: In simulation (Fig. 2d), experiment (Fig. 3e), earth's magnetotail (Fig. 4b), the electron's density has been plotted. However, for heliospheric current sheet, the proton's density is presented. It would be good to know if there is any particular reason.

6. line [531-532]: The authors have neglected the pressure term in equation (1), by saying that it contributes only $\sim$10% to the in-plane electric field. At least any physical intuition and/or reference would be nice.

7. line [545-547]: Equation (4) should have a '-'ve sign. Also, n_{av} is not defined.

8. line [572-573]: Why is the velocity required for converting E_x to potential?

9. line [789-792]: The equilibrium field configuration in equations (7) and (8) is not correctly written. For example, the argument of tanh should have x, not z. Please check them thoroughly. L_{pert} is not described anywhere in the text.

10. line [866-869]: Please check the simulation domain to mimic the MRX experiment in Fig. 9. It is mentioned that $L_r = 30d_i$ and $L_z = 60d_i$. However, in Fig. 9a, the domain sizes are quite smaller. Nevertheless, the separation of flux cores of $18d_i$ matches well with Fig. 9a.

Review Report

We thank the authors for addressing and responding to all of our queries. However, we still have a couple of concerns that prevent us from recommending publication of this manuscript.

Below are the main issues.

1. The authors in fact have not even defined what EBCS actually are. In the references they cite [7, 11-15], the EBCS can extend up to hundreds or thousands of d_i. However, in their Figures 2 (simulation) and 3 (experiment), what they call EBCS only extend up to several d_i. If several d_i extension is sufficient for EBCS, then the current sheet bifurcation in Figure 8 also extends from 0-6 d_i and therefore is an EBCS. But they present Figure 8 as a regime where EBCS does not exist. Therefore, the definition of EBCS should be given clearly as currently it is very ambiguous, and calling a bifurcated current that extends only several d_i as EBCS is questionable.

2. Thanks to the authors' explanations, now we understand what Equation 5 means. Let us first point out two minor points. First, it should be made clear that the electron pressure in Eq. 1 is the local electron pressure but the beta in Equation 5 refers to the global beta. Second, the beta should be properly defined: is it the electron beta, the ion beta, or total (the ratios are all different in the simulations). Now the main point: the expression in Equation 5 tells you nothing about the length scale of the potential drop. This is important because the existence of a large potential drop alone does not explain the localized nature of the current sheet and the electric field. The localized current on each side of the bifurcation signifies a localized Hall field, which means the potential drop must happen over a short distance. Why it occurs over such short distance is not addressed by Eq. 5 and its derivation. If the same potential drop occurs over a much larger spatial distance, such localized current sheets would not arise. Therefore, the present analytical derivation does not address the origin of EBCS.

3. If the authors wish to make the argument that the formation of EBCS depends on the plasma beta, they should perform comparisons between systematically varied simulations. Right now, the four PIC simulations, including the one in the Appendix, all have differing domains, mass ratios, temperature ratios, temperature, sheet widths, etc. The length scales in electron skin depths, which are the species that control the EM field (e.g., Eq. 1) are all over the place. There isn't a single set of two simulations with only beta as the different parameter. It is thus impossible to isolate the cause of the EBCS through these simulations. Also, the plasma beta for the simulation in Figure 2 is not even mentioned, and from Reference 27, we only guess it to be 0.03. This is similar to the simulation in Figure 8, which the authors show as an example of absence of EBCS (this point is also related to point 1 above).

Overall, while the topic is interesting and relevant, and there indeed seems to be some quality material, the current analysis does not convincingly support the central claim regarding expanding bifurcated current sheets. A more controlled parameter study, a clarification of the variables, and a more concrete explanation of the origins of EBCS are absolutely necessary in order for this work to be scientifically accurate, persuasive, and meaningful.

One minor point:

The authors state that EBCS emerges robustly during quasi-steady reconnection. However, in their simulations, they do not explicitly demonstrate whether these structures actually appear in the steady state reconnection. Rather, in Figs. 2, 6, and 8, they present only a single snapshot (i.e., at one time) for each case.

The temporal evolution and dynamics of these structures are not shown anywhere in the manuscript, despite being accessible within their simulations. To support their claim, it would be important to explicitly demonstrate the formation and persistence of EBCS during the steady-state phase of reconnection, at least in one representative simulation, perhaps Figure 8.

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